

ZERO KNOWLEDGE PROOF eKYC

A Privacy-Preserving Selectively Disclosed eKYC System using Zero-Knowledge Proofs, Merkle Trees and Post-Quantum Signatures

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ABSTRACT

Traditional electronic Know-Your-Customer (eKYC) frameworks mandate that individuals transmit unredacted, high-resolution scans of government-issued identity documents (such as Indian Aadhaar cards, Passports, and PAN cards) alongside raw photographic selfies to centralized institutional servers. This legacy paradigm creates massive, highly vulnerable centralized data honeypots that are perpetually susceptible to catastrophic data breaches, internal data misuse, identity theft, and synthetic identity fraud. Furthermore, it enforces excessive oversharing: proving an elementary predicate, such as legal adulthood ($\text{Age} \geq 18$) or institutional enrollment, necessitates disclosing full legal names, residential addresses, precise dates of birth, and unique national identity numbers. This report presents the design, mathematical formalization, and implementation of a next-generation Privacy-Preserving, Selectively Disclosed eKYC System built upon Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge (zk-SNARKs) and algebraic Poseidon Merkle Trees.

The proposed architecture adheres to the decentralized Self-Sovereign Identity (SSI) paradigm, decoupling the user's identity verification into a three-party trust model: Issuers, Holders, and Verifiers. Using an arithmetic circuit (37,791 R1CS constraints) written in Circom 2.1.9 over the BN128 elliptic curve and proved using the Groth16 proving system, a citizen can generate mathematical proofs of claims (e.g., age eligibility and credential non-expiry) while keeping sensitive personal identifiers completely confidential. The system introduces several pioneering enterprise-grade innovations: (1) a Hybrid Post-Quantum Signature Layer combining classical in-circuit EdDSA over BabyJubJub with quantum-resistant ML-DSA-65 (NIST FIPS 204 / Dilithium3) signatures verified at wallet import; (2) an On-Device Biometric Liveness and Facial Matching Engine executing entirely within the client browser via WebAssembly and HTML5 Canvas, analyzing temporal RGB pixel flux micro-motion and computing 64-bin gradient orientation cosine similarities without transmitting raw biometric data over the network; (3) a Tri-Registry Government Validation Engine incorporating the Verhoeff D5 dihedral group algorithm for Aadhaar, ICAO Doc 9303 checksums for Passports, and NSDL syntax validation for PAN; (4) a Zero-Knowledge Anti-Money Laundering (zk-AML) Sanctions Screening mechanism utilizing 20-level Sparse Merkle Trees (SMT) for cryptographic non-membership proofs; and (5) Ethereum smart contracts

(EKYCRegistry.sol) providing on-chain verification and single-use nullifiers for front-running and replay defense.

The implementation is released as an open-source repository (github.com/Debugger-NaN/ekyc-zkp) with a command-line toolchain, a browser-based identity vault, a Node.js verifier gateway, Solidity contracts and a scripted multi-party trusted-setup ceremony. Measured on a 2-vCPU Linux container running Node.js 22, a Groth16 proof is generated in about 4 seconds, the proof is under 1 KB, and off-chain verification takes about 35 milliseconds; browser proving on the authors' development workstation takes approximately 1.4 seconds. The end-to-end integration suite (12 tests) and the zk-AML suite (3 tests) pass, covering replay, session-binding, forged post-quantum signature, untrusted-issuer, under-age predicate and wallet-encryption scenarios. The resulting system removes the need for centralised identity honeypots, supports the data-minimisation principles of the GDPR and India's DPDP Act 2023, and provides a quantum-resistant foundation for digital identity in banking, telecommunications, governance and decentralised finance.

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CHAPTER 1: PROJECT DESCRIPTION AND OUTLINE

1.1 Introduction

In today's hyper-connected global economy, digital identification forms the bedrock of modern commercial, legal, and governmental interactions. Know Your Customer (KYC) regulations are legally mandated worldwide across banking, non-banking financial companies (NBFCs), securities exchanges, telecommunications providers, cryptocurrency platforms, and higher educational institutions. KYC processes are designed to combat identity fraud, money laundering, financial terrorism, and tax evasion. However, the conventional mechanisms employed to achieve KYC compliance remain dangerously archaic, centralized, and architecturally misaligned with fundamental human privacy rights.

Under the prevailing KYC paradigm, whenever an individual opens a bank account, obtains a SIM card, registers for an academic degree, or executes a financial transaction, they are required to submit unredacted digital scans (or physical photocopies) of primary government credentials—such as Indian Aadhaar cards, Passports, Permanent Account Number (PAN) cards, or Driver's Licenses—coupled with photographic portrait selfies. These documents contain an exhaustive catalog of highly confidential personal attributes: full legal names, parents' names, residential street addresses, exact dates of birth, unique national identification numbers, biometric markers, and signatures. Once uploaded, these sensitive digital assets are archived in perpetuity across thousands of corporate databases and cloud storage buckets.

This architectural paradigm causes two profound systemic failures: Centralized Data Honeypots and Catastrophic Oversharing. Centralized data repositories aggregate millions of identity dossiers, transforming corporate servers into lucrative targets for domestic and foreign cyber adversaries. Breaches of these repositories expose citizens to irreversible identity theft, blackmail, synthetic identity creation, and unauthorized financial profiling. Concurrently, traditional verification forces total data exposure even when an entity requires only an isolated, boolean predicate. For instance, an online service or retail venue verifying whether a customer is over 18 years old learns their full birth date, home address, and identification number, despite possessing zero legitimate operational need for that contextual metadata.

This project resolves this fundamental crisis by architecting and implementing a Privacy-Preserving, Selectively Disclosed Electronic Know-Your-Customer (eKYC) system utilizing Zero-Knowledge Proofs (zk-SNARKs) and Poseidon Merkle Tree Commitments. Leveraging the Groth16 zk-SNARK protocol over the BN128 elliptic curve, our system enables individuals to cryptographically prove assertions—such as being over a threshold age, holding an active university affiliation, and possessing an unexpired credential from an authorized trust anchor—without disclosing their date of birth, identity card numbers, or underlying cryptographic signatures. By coupling this mathematical foundation with post-quantum hybrid signatures (ML-DSA-65), on-device client-side biometrics, tri-registry government cross-validation, and zero-knowledge sanctions screening, this project delivers an uncompromising, production-grade identity architecture that guarantees absolute data minimization and user sovereignty.

1.2 Motivation for the Work

The motivation driving this research and engineering initiative stems from the urgent confluence of cyber-security vulnerabilities, regulatory mandates, and cryptographic evolution:

- **Epidemic of Identity Breaches:** Global security analyses reveal that identity records and personal credentials account for over 75% of all breached data records worldwide. In India alone, leaks involving national databases and telecom repositories have exposed millions of citizens' Aadhaar numbers and demographic records to unauthorized exploitation. Developing an eKYC paradigm that operates without centralized storage of raw documents is an urgent civil and national security necessity.
- **Regulatory Compliance and Privacy Mandates:** The General Data Protection Regulation (GDPR) enacted by the European Union explicitly codifies Article 5(1)(c) 'Data Minimization' and Article 5(1)(e) 'Storage Limitation'. Similarly, India's Digital Personal Data Protection (DPDP) Act 2023 enforces stringent statutory penalties (up to ₹250 Crores) for organizations that collect non-essential personal information or fail to prevent personal data leaks. Financial Action Task Force (FATF) guidance further advocates for privacy-preserving digital identity systems. Zero-Knowledge Proofs represent the only mathematical construct capable of achieving regulatory compliance while preserving absolute privacy.

- **The Threat of the Post-Quantum Era:** Mainstream public-key infrastructure—such as RSA, ECDSA, and classical Ed25519—relies on the computational hardness of prime factorization and discrete logarithms. Advances in quantum computing (specifically Shor's algorithm) will inevitably render these primitives obsolete within the next decade. Identity credentials issued today have multi-decade validity horizons; thus, embedding Post-Quantum Cryptography (PQC) into verifiable credentials is a non-negotiable imperative.
- **Biometric Sovereignty:** Traditional biometric verification platforms upload raw biometric imagery (facial scans, fingerprint templates) to third-party cloud APIs, where they risk interception and compromise. Biometric characteristics cannot be changed once compromised. Consequently, a strong motivation of this project is to pioneer an On-Device Biometric Engine that executes 100% locally in client-side WebAssembly, ensuring that raw biometric vectors never cross network boundaries.

1.3 Problem Statement

Conventional eKYC infrastructures impose an unacceptable tradeoff between regulatory compliance and personal privacy, characterized by the following critical flaws:

- **Structural Over-Exposure:** Verification relies on transmitting complete document images, exposing irrelevant attributes (e.g., residential address and national ID number exposed during simple age verification).
- **Single-Point-of-Failure Centralization:** Relying parties store unencrypted or symmetrically encrypted documents, generating massive honeypots vulnerable to external penetration and rogue insider exfiltration.
- **Replay Vulnerability and Cross-Context Linkability:** Stolen credential PDFs or static digital signatures can be re-submitted maliciously across different verifiers, while persistent identifiers enable surveillance networks to correlate user activity across independent service providers.
- **Quantum Cryptanalytic Fragility:** Existing verifiable identity platforms rely solely on pre-quantum elliptic curve algorithms that are vulnerable to quantum cryptanalysis.

Problem Formulation: To design, implement, and mathematically validate a decentralized, privacy-preserving eKYC framework utilizing zero-knowledge succinct non-interactive arguments of knowledge (zk-SNARKs), Poseidon Merkle trees, and

hybrid post-quantum cryptography, enabling citizens to selectively disclose verified attributes and prove predicate claims without disclosing private demographic data, transmitting biometric markers, or trusting centralized document repositories.

1.4 Objective of the Work

The primary objective of this project is to construct a production-ready, mathematically rigorous, privacy-preserving eKYC platform. The specific operational and engineering objectives are as follows:

- **Circuit Architecture:** Design and compile an arithmetic constraint circuit using Circom 2.1.9 that verifies a 16-leaf Poseidon Merkle tree, validates issuer EdDSA signatures in-circuit, enforces 32-bit range checks for age ($\text{Age} \geq \text{minAge}$) and credential expiry without modular field wrapping, and computes deterministic nullifiers.
- **Multi-Party Trusted Setup:** Script and document an enterprise-grade Groth16 Multi-Party Computation (MPC) ceremony protocol incorporating Powers of Tau and circuit-specific Phase-2 contributions to eliminate toxic-waste trust assumptions.
- **Tri-Registry Integration:** Develop an authoritative government validation engine cross-referencing Indian Aadhaar (with Verhoeff D5 dihedral group checksums), PAN (Income Tax/NSDL syntax validation), and Passports (ICAO Doc 9303 TD3 Machine-Readable Zone modulus-10 checksums) with word-order-insensitive name normalization.
- **On-Device Biometric Engine:** Engineer an in-browser WebAssembly/Canvas biometric engine providing interactive anti-spoofing liveness verification via frame-to-frame pixel flux analysis (1.2%–45% threshold) and 64-bin Histogram of Oriented Gradients (HOG) facial cosine matching ($\geq 65\%$ threshold).
- **Zero-Knowledge Sanctions (zk-AML) Screening:** Implement a 20-level Sparse Merkle Tree (SMT) representing international sanctions watchlists (OFAC/PEP), enabling clean citizens to mathematically prove non-membership without revealing their identity.
- **Post-Quantum Hybrid Security:** Integrate dual digital signatures combining classical BabyJubJub EdDSA for in-circuit Groth16 efficiency with NIST FIPS 204 ML-DSA-65 (Dilithium3) lattice signatures verified at wallet import.

- **On-Chain Smart Contract Verification:** Deploy EVM-compatible Solidity smart contracts (EKYCRegistry.sol and Groth16Verifier.sol) executing pairing checks over BN128, civil date synchronization, and single-use nullifier tracking to prevent front-running and replay attacks.

1.5 Scope and Source Repository

The complete implementation accompanying this report is maintained in the public GitHub repository **Debugger-NaN/ekyc-zkp** (<https://github.com/Debugger-NaN/ekyc-zkp>). It is a Node.js project (ekyc-zkp v1.0.0) that bundles every layer of the system: the Circom circuit and its pre-built proving artefacts (build/), the issuer, holder and verifier command-line tools (scripts/), the business-logic modules (src/), the HTTP gateway (gateway/), the browser identity vault (wallet/), the Solidity contracts (contracts/), legacy-integration samples (legacy/) and the trusted-setup ceremony scripts (scripts/ceremony/). The directory layout is given in Appendix C and the run instructions in Appendix D.

The project extends the baseline scheme of Ahmed et al. (APCC 2025), which demonstrated selective disclosure with a Merkle tree and zero-knowledge proofs, with five additions: hybrid post-quantum signatures, an on-device biometric engine, tri-registry document authentication, zero-knowledge sanctions screening and on-chain verification. The government registries, the sanctions list and the issuer are **simulated** with local JSON data; integration with live UIDAI, NSDL or passport services is planned for Phase-II.

1.6 Summary

This chapter established the foundational framework of the Capstone Project. It analyzed the severe structural vulnerabilities of legacy centralized KYC paradigms—principally honeypot risks and systemic oversharing—and articulated the motivation for building a Zero-Knowledge eKYC architecture. The formal problem statement and enumerated technical objectives define a clear path toward developing a multi-layered identity platform combining zk-SNARKs, post-quantum cryptography, client-side biometrics, and tri-registry validation. Subsequent chapters provide detailed investigations into related works, technical requirements, architectural design, implementation benchmarks, and real-world applicability.

CHAPTER 2: RELATED WORK INVESTIGATION

2.1 Existing Approaches and Methods

The evolution of identity verification architectures encompasses several distinct technological paradigms, spanning centralized databases, federated identity providers, decentralized identity frameworks, and zero-knowledge protocols:

1. Centralized e-KYC and Cloud Identity Verification SaaS

The dominant model globally relies on centralized identity brokers and commercial verification vendors. In India, the UIDAI Aadhaar e-KYC platform allows licensed entities (AUAs/KUAs) to authenticate citizens via One-Time Passwords (OTP) or biometric fingerprint/iris scans. Upon authentication, the UIDAI server returns an XML dossier containing the citizen's full demographic record and photograph. Similarly, commercial cloud platforms (such as Jumio, Onfido, and Trulioo) require users to capture live video selfies and upload document scans, which are processed using proprietary computer vision and optical character recognition (OCR) models hosted in centralized corporate clouds. While legally established, these architectures suffer from extreme centralization: every verification event is logged centrally, enabling citizen surveillance, and massive repositories of personal data are exposed to data breach risks.

2. Federated Identity Frameworks (OAuth 2.0 & OpenID Connect)

Federated identity models (such as Google Sign-In, Apple ID, and OpenID Connect) streamline authentication across web services by delegating credentials to centralized Identity Providers (IdPs). The relying party receives signed JSON Web Tokens (JWTs) containing identity claims. However, federated identity was designed for authorization and access control rather than legally compliant KYC verification. Furthermore, federated IdPs can track every service the user logs into, creating severe cross-site profiling and vendor lock-in concerns.

3. Self-Sovereign Identity (SSI) with W3C Verifiable Credentials

The Self-Sovereign Identity (SSI) movement, standardized by the World Wide Web Consortium (W3C), introduced Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs). Frameworks such as Hyperledger Indy, Aries, and Sovrin allow issuers to sign credentials using public-key cryptography and deliver them to user-held digital wallets. While W3C VCs decouple verification from continuous issuer availability,

classical implementations rely on standard digital signatures (e.g., ECDSA or Ed25519) over complete credential objects. To prove an attribute, the user must present the signed credential, thereby disclosing all other bundled attributes unless complex, computationally heavy BBS+ signature schemes are deployed.

4. Emerging Zero-Knowledge Identity Systems

Recent cryptographic breakthroughs have spurred the creation of zero-knowledge identity prototypes: (a) Polygon ID utilizes Iden3 Circom circuits and BabyJubJub signatures to prove atomic claims, but lacks integrated multi-government registry validation and post-quantum layers; (b) Anon Aadhaar uses RSA signature verification inside SNARK circuits to prove possession of an Aadhaar card, but requires massive constraint counts (~500,000 constraints) due to SHA-256 and 2048-bit RSA math, resulting in high prover latency on mobile devices; (c) Worldcoin utilizes custom biometric hardware ('the Orb') to generate IrisHash commitments, but faces widespread international regulatory scrutiny and ethical concerns regarding proprietary hardware capture; and (d) Foundational academic research by Ahmed et al. (APCC 2025), entitled 'A Privacy-Preserving Selectively Disclosed eKYC System Using Merkle Tree and Zero-Knowledge Proofs', which demonstrated the feasibility of Poseidon-based Merkle trees for attribute disclosure, serving as the core conceptual baseline extended in our project.

2.2 Pros and Cons of the Stated Approaches and Methods

To rigorously evaluate the state of the art, Table 2.1 provides an exhaustive multi-dimensional comparison of existing identity paradigms against our proposed Zero-Knowledge eKYC system.

Table 2.1: Comparison of identity paradigms with the proposed ZK-eKYC system

Feature / Paradigm	Centralized eKYC (UIDAI / Jumio)	Federated Identity (OIDC / OAuth)	W3C VCs (Hyperledger Indy)	Proposed ZK-eKYC System
Data Transmission	Full unredacted document & selfie	Signed JWT with identity claims	Signed credential JSON payload	Zero PII; 800-byte zk-SNARK proof
PII Exposure	Total (Name, DOB, Address, NID)	Partial to High (Email, Profile)	Total to Verifier upon display	Zero; only boolean predicates proven
Server Storage Risk	Severe (Centralized Honeypot)	High (IdP stores user dossier)	Low (Stored in user wallet)	Zero (Verifier stores nullifier hash)
Biometric Privacy	Raw video/selfie sent to cloud	None / Delegated to platform	Externalized / Out-of-band	100% on-device WebAssembly/Canvas
Revocation Privacy	Centralized DB lookup leaks use	IdP checks revocation online	Cryptographic Accumulator/RevList	20-Level SMT Non-Membership Proof
AML / Sanctions	Centralized plaintext name scan	Centralized watchlist query	Manual / Out-of-band check	Zero-Knowledge Sanctions SMT check

Feature / Paradigm	Centralized eKYC (UIDAI / Jumio)	Federated Identity (OIDC / OAuth)	W3C VCs (Hyperledger Indy)	Proposed ZK-eKYC System
Replay Defense	Static documents easily copied	Session nonces / Short-lived tokens	Challenge-response signatures	Cryptographic Poseidon nullifiers
Post-Quantum Ready	No (Vulnerable RSA / ECDSA)	No (RSA / ECDSA tokens)	No (Classical elliptic curves)	Yes (Hybrid EdDSA + ML-DSA-65)

Analytical Discussion: As demonstrated in Table 2.1, centralized eKYC solutions prioritize institutional convenience at the complete expense of user privacy and systemic resilience. While federated identity and basic W3C Verifiable Credentials improve decentralization, they remain vulnerable to attribute oversharing, correlation attacks, and post-quantum cryptanalysis. In contrast, our proposed architecture synergizes zk-SNARKs, algebraic Poseidon hashing, sparse Merkle tree non-membership, on-device biometric anti-spoofing, and NIST-standardized post-quantum signatures, achieving an unprecedented balance of security, efficiency, regulatory compliance, and individual sovereignty.

2.3 Research Gap

The surveyed systems each solve part of the problem. Centralised eKYC and federated identity maximise convenience but concentrate personal data; W3C Verifiable Credentials move storage to the user but still reveal whole attributes unless heavier BBS+ signatures are used; Anon Aadhaar proves possession of an Aadhaar card but needs on the order of 500,000 constraints for in-circuit RSA and SHA-256; and Polygon ID provides atomic claims but has no document-checksum validation, sanctions screening or post-quantum protection. No surveyed system combines (i) a small Poseidon-based circuit, (ii) hybrid classical and post-quantum issuer signatures, (iii) privacy-preserving revocation and sanctions checks, (iv) on-device biometrics and (v) on-chain, front-running-resistant verification. This project targets that combination.

CHAPTER 3: REQUIREMENT ARTIFACTS

3.1 Introduction

Requirements engineering for zero-knowledge cryptographic systems requires rigorous specification across mathematical, structural, performance, and security dimensions. Unlike traditional software systems where runtime errors can be resolved via procedural exception handling, zero-knowledge arithmetic circuits operate over algebraic prime fields where any under-constrained signal, unhandled modulo wrap, or missing assertion introduces fatal soundness or zero-knowledge vulnerabilities. This chapter establishes the comprehensive requirement artifacts governing the development, deployment, and verification of our eKYC system.

3.2 Hardware and Software Requirements

3.2.1 Hardware Requirements

- **Client / Holder device:** consumer desktop, laptop or mobile workstation with at least 4 GB RAM, a dual-core 2.0 GHz x86-64 or ARM64 processor and a 720p webcam (30 fps) for on-device biometric capture.
- **Issuer and gateway node:** dual-core server with 8 GB RAM and 20 GB SSD capable of hosting the Node.js REST/JSON/XML gateway and serving static cryptographic artefacts (about 30 MB).
- **Ethereum node / EVM network:** local development network (Hardhat or Anvil) or a public testnet such as Sepolia.

3.2.2 Software Requirements

Table 3.1: Software requirements and dependency versions

Component	Requirement (as used in the repository)
Runtime	Node.js v20 or higher (v22 tested); npm v9 or higher; OpenSSL on PATH
ZK toolchain	Circom 2.1.9 (only to recompile); snarkjs ^0.7.6; circomlib ^2.0.5; circomlibjs ^0.1.7 (Poseidon, EdDSA, BabyJubJub, SMT)
Post-quantum	@noble/post-quantum ^0.7.1 (ML-DSA-65, FIPS 204)
Smart contracts	Solidity 0.8.x (solc ^0.8.24); EVM precompiles ecAdd (0x06), ecMul (0x07), ecPairing (0x08) on alt_bn128
Browser vault	Chrome, Firefox or Edge 115+ with WebAssembly, Web Workers, Web Crypto, IndexedDB/localStorage and Canvas
OCR / parsing	tesseract.js ^7.0.0 and pdfjs-dist ^6.3 (bundled locally under wallet/vendor)
Cryptography (vault)	Web Crypto PBKDF2 (250,000 iterations, SHA-256) and AES-256-GCM

3.3 Specific Project Requirements

3.3.1 Data Requirements

The system mandates precise structuring of identity attributes within prime field arithmetic over the BN128 scalar field ($r = 21888242871839275222246405745257275088548364400416034343698204186575808495617$):

- **Attribute schema:** nine attribute slots are mapped into a depth-4 Merkle tree of 16 leaves. Slots 0–8 hold Name, Date of Birth, Nationality, National ID, Address, University, Holder Commitment, Expiry Date and Credential ID respectively; slots 9–15 are zero padding.
- **Salt commitments:** every attribute is masked by a unique 256-bit random salt, $\text{Leaf}_i = \text{Poseidon}(\text{Value}_i, \text{Salt}_i)$, which defeats dictionary brute-force attacks.
- **Temporal formats:** all dates (DOB, expiry, current date) are positive 32-bit integers in YYYYMMDD form (for example 5 October 2026 is 20261005).
- **Nullifier:** a deterministic single-use public signal $\text{Nullifier} = \text{Poseidon}(\text{holderSecret}, \text{verifierId}, \text{nonce})$.

3.3.2 Functional Requirements

- **Document Ingestion & Checksum Validation:** The system must ingest raw documents, extract text via in-browser OCR, dynamically detect document type (Aadhaar, Passport, PAN), and execute mathematical checksum verification (Verhoeff D5 for Aadhaar, ICAO 9303 for Passport, syntax regex for PAN).
- **Institutional Credential Issuance:** The issuer must verify applicant details against authoritative government databases, construct the 16-leaf Poseidon Merkle tree, and sign the Merkle root using dual digital signatures: EdDSA over BabyJubJub and ML-DSA-65.
- **On-Device Biometric Verification:** The browser wallet must execute anti-spoofing liveness verification via temporal RGB pixel flux analysis across sequential video frames, extract 64-bin HOG feature vectors, and compute cosine similarity against the document portrait.

- **zk-AML Sanctions Screening:** The system must compile international watchlists into a 20-level Sparse Merkle Tree (SMT) and enable unsanctioned users to generate non-membership proofs without disclosing identity.
- **Selective Disclosure Proof Generation:** The client prover must calculate Groth16 witness and proof files entirely in WebAssembly within the browser.
- **8-Point Verifier Audit:** The relying party must execute an 8-point audit checklist verifying the SNARK proof, issuer trust anchor, clock drift, policy parameters, verifier ID, session nonce, revocation root, and nullifier uniqueness.

3.3.3 Performance and Security Requirements

- **Constraint budget:** the circuit shall stay below 50,000 R1CS constraints so that proving remains practical in a browser.
- **Latency:** proof generation under 5 seconds on a 2-vCPU machine (target about 2 seconds on a modern desktop); verifier latency under 50 milliseconds.
- **Cryptographic security:** about 128-bit classical security on BN128 and NIST security category 3 for ML-DSA-65.
- **Zero-knowledge privacy:** the proof and public signals reveal nothing about hidden attributes beyond the proven predicates.
- **Replay defence:** single-use nullifiers cause any re-submitted proof to be rejected.

3.3.4 Look and Feel Requirements

- **Responsive Web Vault:** Clean, dark/light theme single-page web application featuring clear step-by-step progress bars (Document Import -> Biometrics -> Sanctions -> Issuance -> Proof Generation).
- **Transparent Consent Modal:** Intuitive consent dialogue explicitly displaying which attributes are shared publicly (e.g. Name, University) versus which attributes remain 100% cryptographically hidden (e.g. DOB, National ID, Address).
- **Visual Viewfinder & Feedback:** Interactive camera viewfinder with facial bounding guides and dynamic real-time liveness instructions.

3.4 Summary

This chapter established the rigorous requirement artifacts governing the eKYC platform. It detailed hardware and software specifications, prime field data schemas, functional verification workflows, constraint budgets, and user experience requirements. These

artifacts form the technical foundation for the design methodology and system architecture elaborated in Chapter 4.

CHAPTER 4: DESIGN METHODOLOGY AND ITS NOVELTY

4.1 Methodology and Goal

The engineering methodology adopted for this project synthesizes User-Centered Design (UCD) with rigorous Cryptographic Protocol Engineering. Given that zero-knowledge architectures shift the burden of proof generation to the user's local hardware, the primary goal was to construct a system that achieves maximum mathematical security, post-quantum resilience, and regulatory compliance while maintaining sub-2-second consumer-grade responsiveness on ordinary web browsers.

The development lifecycle followed an iterative Agile development model consisting of five phases: (1) Mathematical Modeling and Circuit Constraint Formulation; (2) Cryptographic Prototyping and SMT Revocation Structuring; (3) Post-Quantum Hybrid Integration; (4) Client-Side WebAssembly Vault & On-Device Biometric Engineering; and (5) Gateway Integration, Smart Contract Deployment, and Automated End-to-End Auditing.

4.2 Functional Modules Design and Analysis

The system is decomposed into five functional modules. File locations refer to the repository in Appendix C.

- **Module 1 – Tri-registry document authentication** (`src/registry.js`, `wallet/doc-import.mjs`, `gateway/server.js`): extracts fields by OCR, validates the Verhoeff D5 check digit for Aadhaar, the ICAO Doc 9303 modulus-10 check digits of the passport MRZ and the PAN syntax, matches the record against the simulated UIDAI, MEA and NSDL registries (`data/government-registry.json`) with word-order-insensitive name normalisation, and returns generic errors that never leak the registered name.
- **Module 2 – Issuer and hybrid signer** (`src/issuer-core.js`, `src/crypto-suite.js`): checks `holderCommit = Poseidon(holderSecret)`, draws 256-bit salts, builds the 16-leaf Poseidon Merkle tree, and signs the root with BabyJubJub EdDSA-Poseidon (verified in-circuit) and ML-DSA-65 (verified by the wallet at import).

- **Module 3 – Client-side vault and biometrics** (`src/wallet-core.mjs`, `wallet/biometrics.mjs`): stores credentials encrypted with PBKDF2-derived AES-256-GCM keys and runs the webcam liveness and face-match checks entirely in the browser.
- **Module 4 – Zero-knowledge proving engine** (`circuits/selective_disclosure.circom`): the Circom 2.1.9 circuit that checks the issuer signature, Merkle inclusions, age and expiry bounds, SMT non-revocation and derives the nullifier.
- **Module 5 – Verifier gateway and on-chain registry** (`gateway/server.js`, `src/verify-core.js`, `contracts/EKYCRegistry.sol`): REST/JSON/XML endpoints, the 8-point audit, zk-AML proof checking, and on-chain Groth16 verification with nullifier tracking.

4.3 Software Architectural Designs

4.3.1 Three-Party Self-Sovereign Identity Trust Model

The system follows the decentralised Self-Sovereign Identity (SSI) triangle of Issuer, Holder and Verifier (Figure 4.1):

- **Trust Anchor (Issuer):** Authoritative government agency, passport office, or university that verifies legal documentation, constructs the salted Merkle tree, and signs the root. The issuer never learns the user's private `holderSecret`.
- **Holder / Prover (Citizen):** The citizen storing their credentials in an encrypted browser vault, generating cryptographic proofs on their own device, and exercising selective disclosure.
- **Relying Party (Verifier):** Commercial bank, telecom operator, employer, or smart contract that issues a challenge (nonce, policy parameters) and cryptographically audits the submitted zk-SNARK proof.

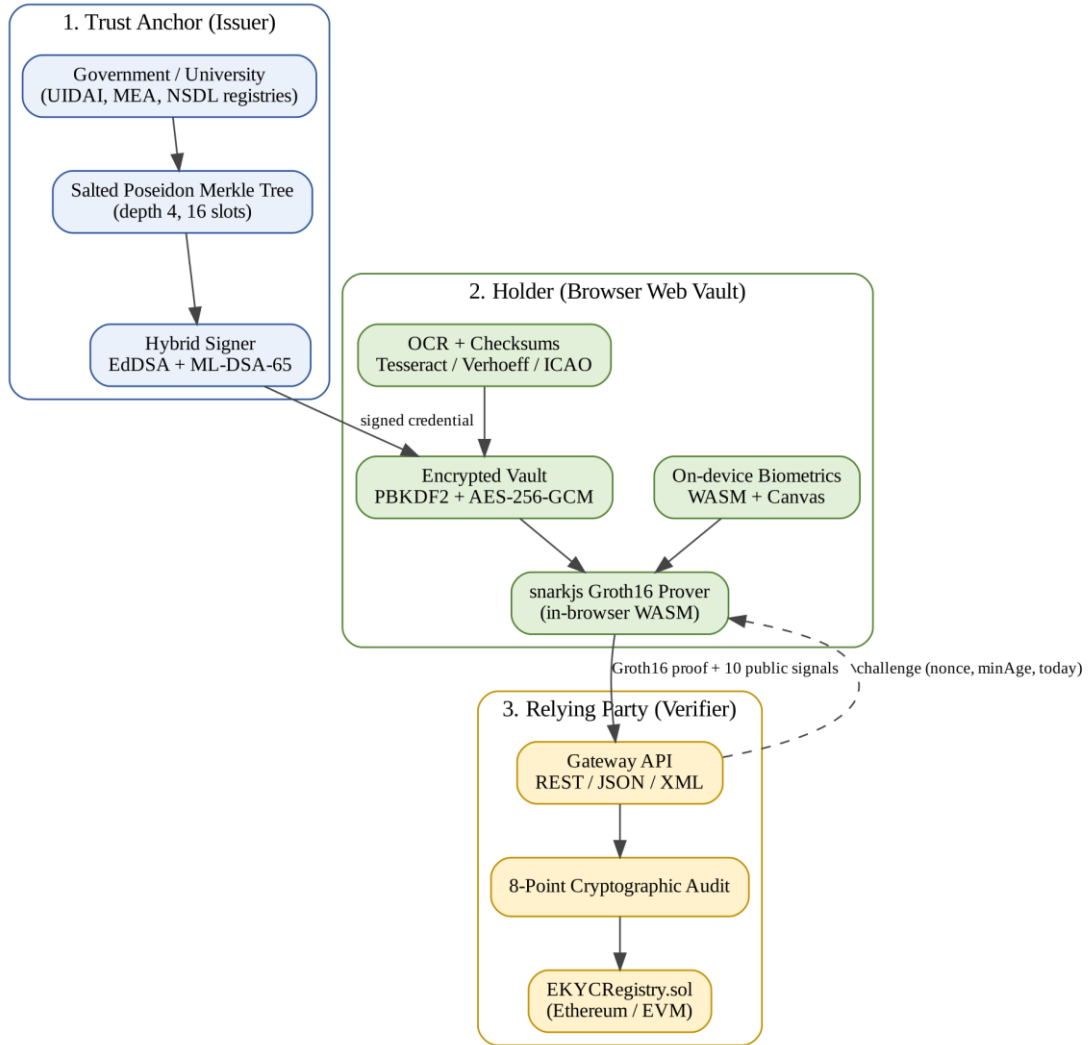


Figure 4.1: Decentralized SSI three-party architecture (Issuer, Holder, Verifier)

4.3.2 End-to-End Lifecycle

Figure 4.2 summarises the nine-step lifecycle implemented by the wallet, issuer and gateway: holder-secret generation and commitment, document ingestion, issuance, credential import with ML-DSA-65 verification, challenge, revocation and sanctions proofs, biometrics and consent, in-browser proving, and verification with nullifier recording.

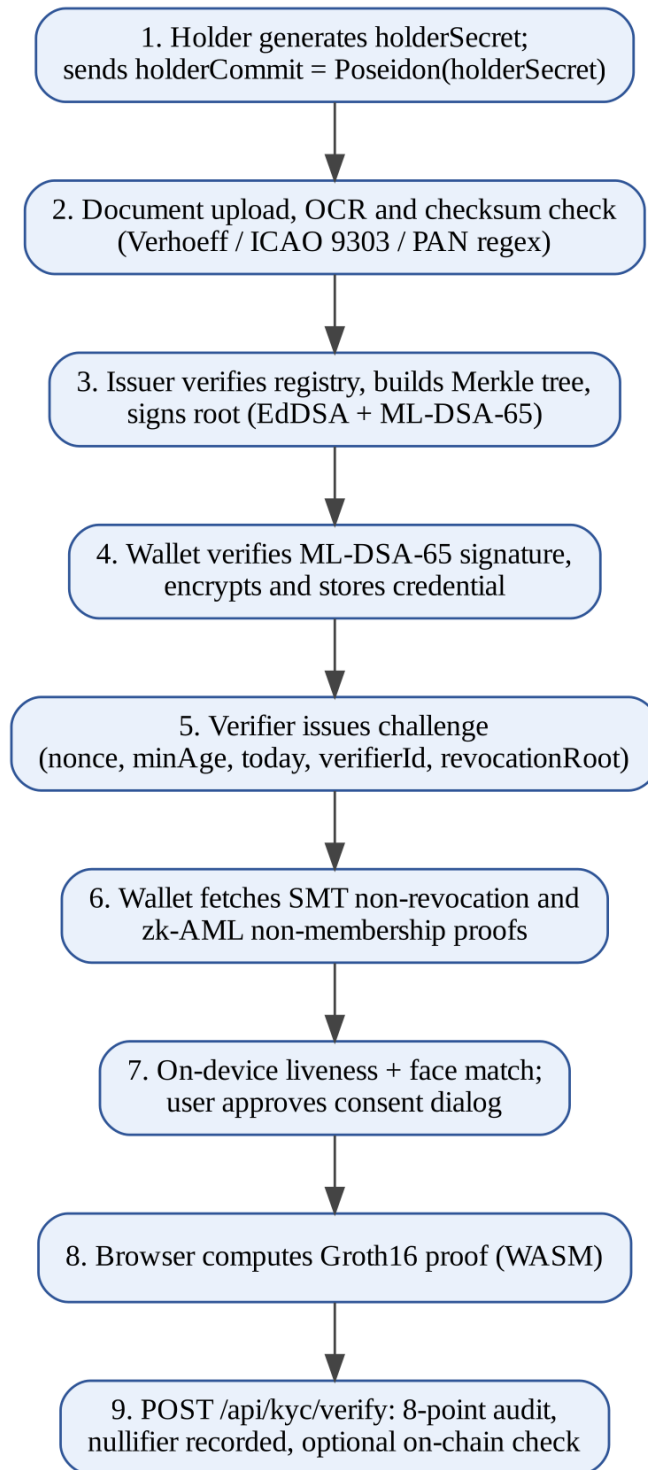


Figure 4.2: End-to-end cryptographic identity lifecycle and verification sequence

4.3.3 Poseidon Merkle Tree (Depth 4)

The identity credential organizes attributes into a fixed-slot binary Merkle Tree of depth 4 (16 leaves, Figure 4.3). Fixed slot assignments allow the circuit to use constant path

indices rather than expensive dynamic witness multiplexers, saving thousands of circuit constraints:

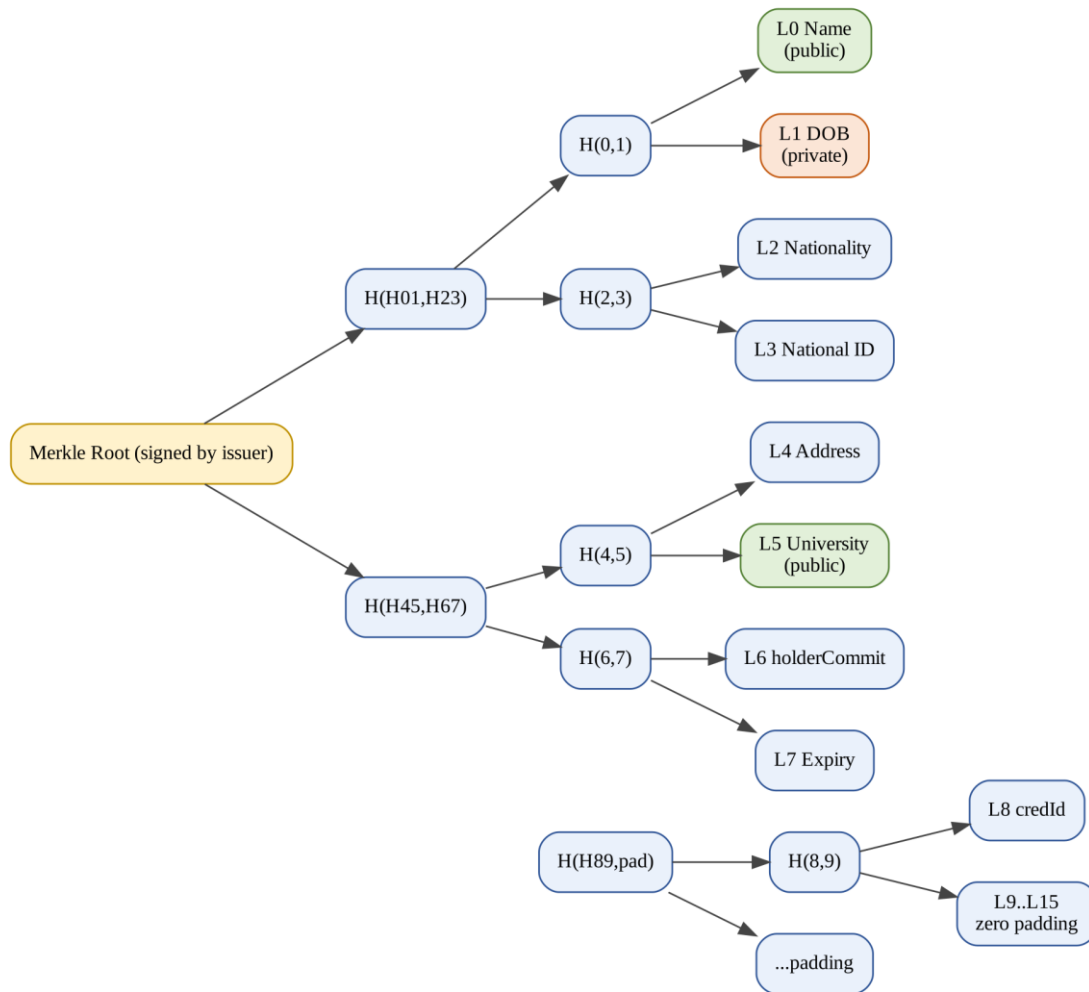


Figure 4.3: Poseidon Merkle tree (depth 4, 16 slots) with salted leaf commitments

Table 4.1: Merkle slot assignment and visibility

Slot	Attribute	Circuit constant	Visibility	Description
0	name	IDX_NAME	Public (if requested)	Name converted to a field element (SHA-256 mod p)
1	dob	IDX_DOB	Private	Date of birth, integer YYYYMMDD
2	nationality	–	Private	ISO-3166 3-letter code
3	nid	–	Private	National ID / passport number
4	address	–	Private	Residential address string
5	university	IDX_UNI	Public (if requested)	University / issuing organisation
6	holderCommit	IDX_HOLDER	Private	Bound to secret: Poseidon(holderSecret)
7	expiry	IDX_EXPIRY	Private	Expiry date, integer YYYYMMDD
8	credId	IDX_CREDID	Private	Unique ID (1..1,048,575), SMT key
9–15	padding	0	Private	Zero leaves completing the tree

4.3.4 Circuit Constraints and Invariants

The circuit `SelectiveDisclosure(4)` has 10 public signals (1 output and 9 inputs) and enforces seven relations (Figure 4.4). Its public-signal order is `[nullifier, issuerAx, issuerAy, today, minAge, nameValue, uniValue, verifierId, nonce, revocationRoot]`.

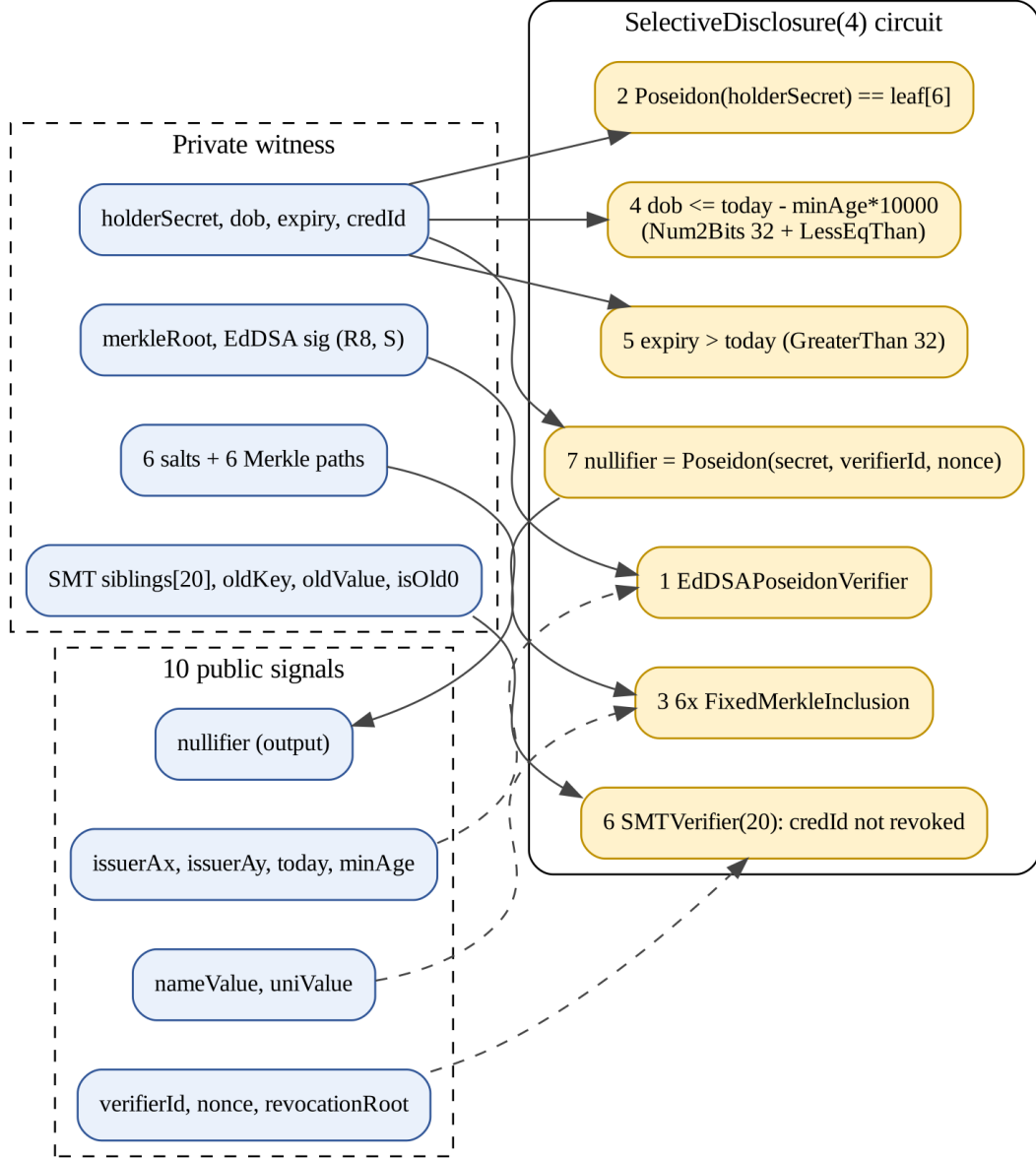


Figure 4.4: Zero-knowledge selective-disclosure circuit signal flow and constraint system

- **Relation 1 (Issuer Signature):** `EdDSAPoseidonVerifier` verifies that the private `merkleRoot` is signed by the public issuer key (`issuerAx`, `issuerAy`).
- **Relation 2 (Holder Binding):** Asserts that `Poseidon(holderSecret) == leaf[6]`. An attacker stealing a credential file cannot generate a valid proof without knowing `holderSecret`.

- **Relation 3 (Merkle Inclusions):** Proves that all 6 verified leaves (name, dob, uni, holderCommit, expiry, credId) correctly hash to the identical private merkleRoot using salted hashes: $\text{Leaf}_i = \text{Poseidon}(\text{Value}_i, \text{Salt}_i)$.
- **Relation 4 (Age Predicate without Field Wrap):** Calculates $\text{threshold} = \text{today} - \text{minAge} * 10000$. Both dob and threshold pass through 32-bit range checks ($\text{Num2Bits}(32)$) and are constrained via $\text{LessEqThan}(32)$ proving $\text{dob} \leq \text{threshold}$.
- **Relation 5 (Credential Validity):** Constrains $\text{expiry} > \text{today}$ using $\text{GreaterThan}(32)$.
- **Relation 6 (Sparse Merkle Tree Non-Revocation):** Proves that credId is absent from the 20-level Sparse Merkle Tree represented by the public revocationRoot.
- **Relation 7 (Replay-Proof Nullifier):** Derives the public output: $\text{Nullifier} = \text{Poseidon}(\text{holderSecret}, \text{verifierId}, \text{nonce})$.

4.3.5 Tri-Registry Government Cross-Referencing

To prevent self-issuance with fake identifiers, the issuer authenticates documents against three simulated authoritative registries before signing (Figure 4.5). Each path combines a mathematical check with a registry lookup:

- **Aadhaar (UIDAI):** Verhoeff D5 check digit on the 12-digit UID, registry lookup, name and date-of-birth match. Errors are `UID_NOT_FOUND`, `NAME_MISMATCH` and `DOB_MISMATCH`.
- **PAN (Income Tax / NSDL):** syntax check $^{[A-Z]\{5\}[0-9]\{4\}[A-Z]\$}$ and registry match.
- **Passport (MEA / Passport Seva):** ICAO Doc 9303 TD3 MRZ check digits using repeating weights 7-3-1, then registry match (`PASSPORT_NOT_FOUND` for unknown numbers).
- **Anti-leak defence:** a name mismatch returns a generic error and never reveals the name on record, preventing fishing attacks.

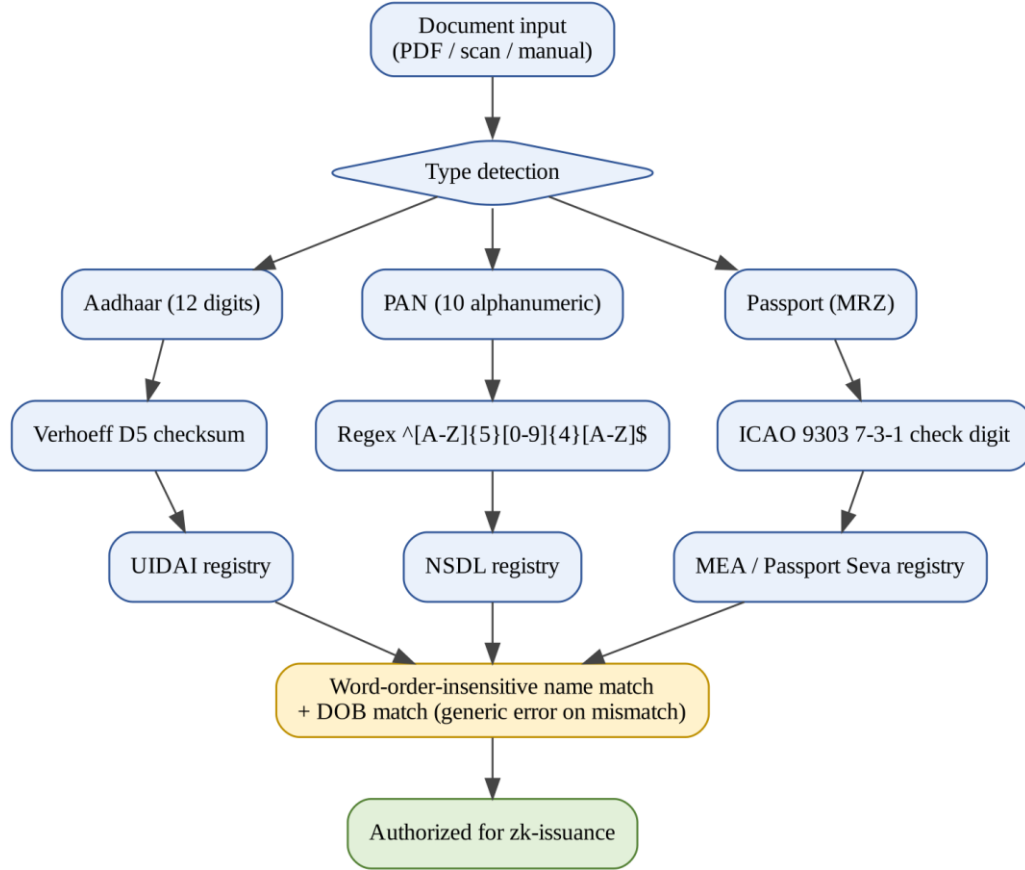


Figure 4.5: Tri-registry multi-government document verification pipeline

4.3.6 On-Device Biometric Engine

The biometric module (`wallet/biometrics.mjs`) runs entirely in the browser (Figure 4.6). Six frames are sampled 300 ms apart (about 1.8 s); the mean frame-to-frame RGB change must fall between 1.2% and 45%, which rejects static photos and screen replays (too little motion) and looping or spliced video (too much). The live face and the document portrait are each reduced to a 64-bin gradient-orientation histogram and compared by cosine similarity,

$$Similarity = (v_{live} \cdot v_{doc}) / (\|v_{live}\| \cdot \|v_{doc}\|), \text{ pass if } Similarity \geq 0.65.$$

Video frames, canvases and feature vectors exist only in volatile browser memory; no biometric data is transmitted.

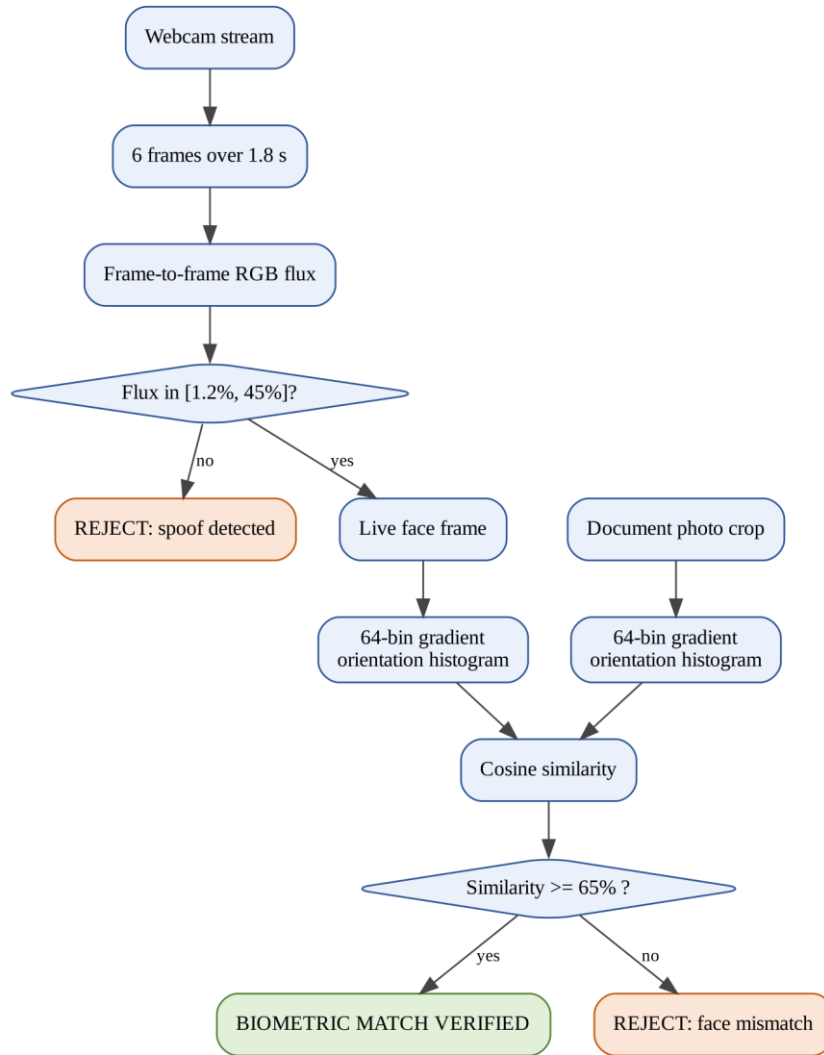


Figure 4.6: On-device biometric anti-spoofing liveness and cosine-similarity flow

4.3.7 Zero-Knowledge AML Sanctions Screening

International sanctions watchlists (OFAC SDN, PEP lists) are compiled into a 20-level Sparse Merkle Tree (SMT). Clean citizens compute an identity commitment key: $\text{Key} = (\text{Poseidon}(\text{NameField}, \text{DOB}) \bmod (2^{20} - 1)) + 1$, and present a cryptographic non-membership proof. Verifiers validate this proof against the authoritative sanctionsRoot, ensuring that sanctioned individuals are blocked without forcing clean citizens to disclose their identities. (Figure 4.7). If a sanctioned person tries to build a proof, the SMT lookup finds the key and an error is raised, so no proof can be produced.

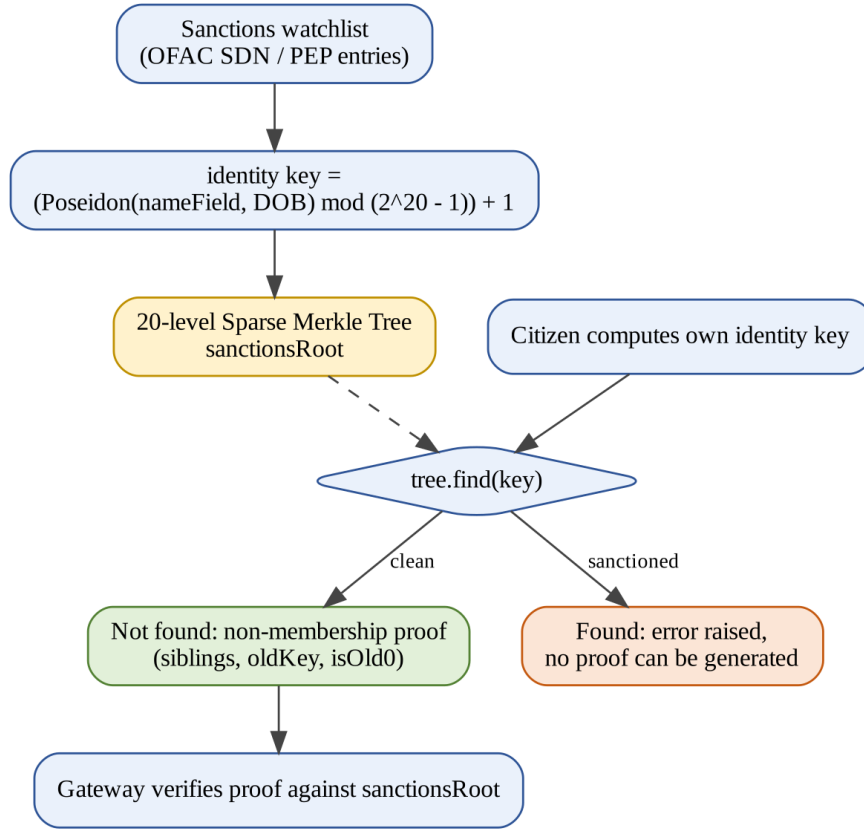


Figure 4.7: zk-AML sparse Merkle tree sanctions non-membership verification flow

4.3.8 Hybrid Post-Quantum Architecture

To safeguard credentials against future quantum computing attacks, the system introduces a hybrid dual-signature paradigm: in-circuit BabyJubJub EdDSA maintains lightweight SNARK proof generation (~1.4s), while NIST FIPS 204 ML-DSA-65 (Dilithium3) lattice signatures protect credential integrity during wallet import. Any attempt to import a forged or tampered credential is instantly rejected before storage. The wallet checks the ML-DSA-65 signature against the issuer public key stored in `data/issuer-public.json`, and the ML-DSA message is domain-separated (`ekyc-root-v1: prefix`) so signatures cannot be replayed in another context. Figure 4.8 shows the two verification paths.

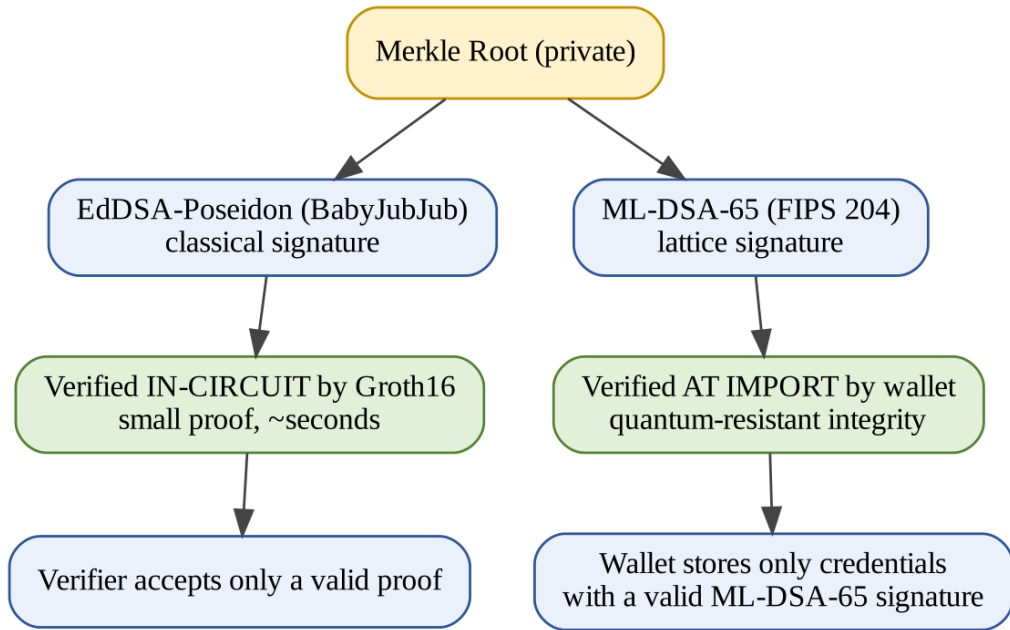


Figure 4.8: Hybrid post-quantum cryptographic architecture (EdDSA + ML-DSA-65)

4.3.9 Multi-Party Trusted Setup Ceremony

Groth16 needs a one-time trusted setup. If one party generated the keys alone, its leftover randomness (“toxic waste”) could be used to forge proofs. The repository therefore scripts a two-phase multi-party ceremony (`scripts/ceremony/` and `CEREMONY.md`): Phase 1 is a circuit-agnostic Powers-of-Tau chain (`phase1_init.sh`, `phase1_contribute.sh`, `phase1_beacon_and_prepare.sh`); Phase 2 is circuit-specific (`phase2_setup.sh`, `phase2_contribute.sh`, `phase2_finalize.sh`) and ends by exporting `circuit_final.zkey`, `verification_key.json` and `Groth16Verifier.sol`. A public random beacon (for example a future Bitcoin block hash or a drand round committed in advance) is applied at the end of each phase, and anyone can audit the transcript with `scripts/ceremony/verify_all.sh` (`npm run ceremony:verify`). The setup is secure as long as at least one contributor is honest and destroys their randomness. The `circuit_final.zkey` shipped in the repository is a development setup; a production deployment must run the full ceremony (Figure 4.9).

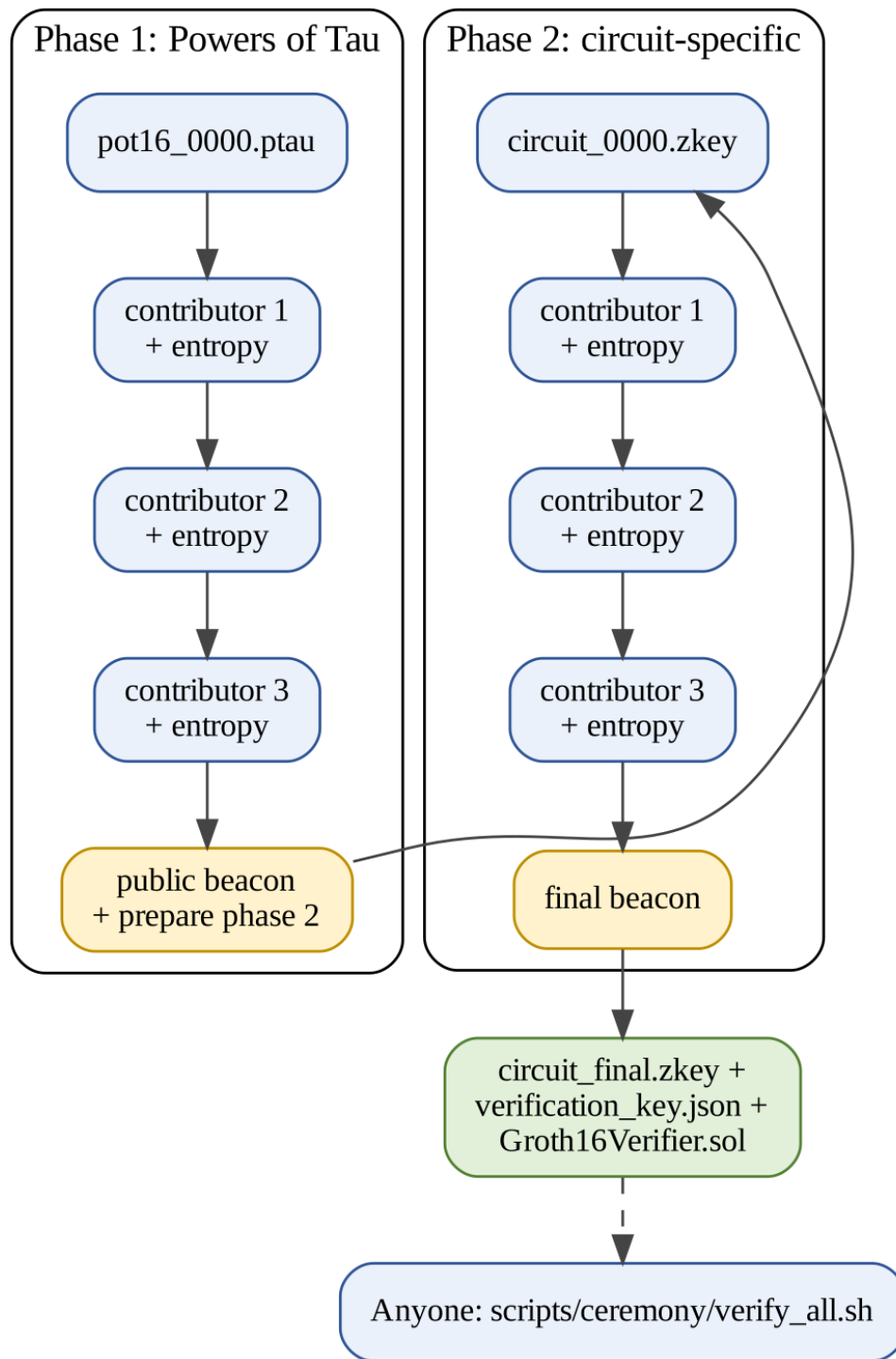


Figure 4.9: Groth16 multi-party trusted setup ceremony (Phase 1 and Phase 2)

4.4 Security and Threat Model

Table 4.2 maps the principal attacks considered in the design to the mechanism that stops each of them.

Table 4.2: Threat model and countermeasures

Threat	Attack scenario	Defence in this system
Replay	Captured proof re-submitted to the same or another verifier	Public nonce and verifierId are bound into the single-use nullifier Poseidon(holderSecret, verifierId, nonce); a repeated nullifier or wrong nonce is rejected (checks 6 and 8).
Credential theft / transfer	Alice gives her credential file to Bob	The circuit needs the preimage holderSecret of leaf 6; the file alone is useless.
Self-issuance	Attacker issues a fake “25 years old” credential with own keys	Proof’s (issuerAx, issuerAy) must equal the trust anchor (data/issuer-public.json or on-chain issuerKeys); otherwise rejected (check 2).
Front-running	Mempool observer re-submits a user’s on-chain proof	EKYCRegistry requires pub[7] == uint256(uint160(msg.sender)); a different caller fails.
Biometric spoofing	Photo, replayed video or avatar held to the camera	Temporal flux window [1.2%, 45%] rejects static and looped media; interactive prompts are required.
Forged post-quantum signature	Tampered ML-DSA-65 signature on import	Wallet verifies ML-DSA-65 before storage; tampered signature rejected (e2e test).
Revoked credential	Holder presents a revoked credId	Wallet requests an SMT non-membership proof; revoked ids get HTTP 403 and no valid witness exists.
Sanctioned identity	Listed person proves eligibility	SMT lookup finds the key; proof generation fails.

4.5 User Interface Designs

The identity vault (`wallet/index.html`, served by the gateway at `/wallet`) is a single-page application whose journeys are:

- **Journey 1 (Document Ingestion):** Drag-and-drop file uploader supporting Aadhaar, Passport, and PAN documents. Live WebAssembly OCR parses demographic fields and validates check digits with real-time feedback.
- **Journey 2 (On-Device Biometrics):** Interactive camera viewfinder guiding the user through liveness prompts ('Look at camera', 'Blink naturally') with real-time motion flux gauges.
- **Journey 3 (Selective Disclosure Consent Modal):** Transparent permission dialogue presenting an itemized matrix of what the verifier will see (e.g. Name: Ahmed, University: OMU, Age ≥ 18 : TRUE) versus what remains hidden (DOB: HIDDEN, Address: HIDDEN, National ID: HIDDEN).
- **Journey 4 (Cryptographic Audit Inspector):** Live proof generator and verifier display presenting SNARK proof parameters, public signals, and mathematical verification status.

Screenshots of the vault served by the running prototype are given in Figure 5.2 and Appendix A.

4.6 Summary

This chapter elaborated the design methodology and technical architecture of the eKYC system. It defined the Three-Party SSI trust model, the 16-leaf Poseidon Merkle tree structure, the mathematical invariants enforced by the Circom circuit, the hybrid post-quantum cryptographic layer, the zk-AML sanctions engine, and the user interface workflows. Chapter 5 examines the concrete implementation details and performance benchmarking.

CHAPTER 5: TECHNICAL IMPLEMENTATION AND ANALYSIS

5.1 Outline

This chapter details the technical coding, concrete code solutions, implementation challenges, and empirical performance analysis of the complete Zero-Knowledge eKYC system. The codebase encompasses Circom circuits, Node.js cryptographic microservices, client-side WebAssembly modules, browser biometric algorithms, and Solidity smart contracts.

5.2 Technical Coding and Code Solutions

5.2.1 Circom Circuit (circuits/selective_disclosure.circom)

The circuit is written in Circom 2.1.9 over BN128 and compiles to **37,791 R1CS constraints** with 37,829 wires, 61 private inputs, 9 public inputs and 1 output (`snarkjs r1cs info build/selective_disclosure.r1cs`). The constraint cost is dominated by the EdDSA-Poseidon verifier, the 20-level SMT verifier and the six depth-4 Poseidon Merkle inclusions. Key templates are:

- `FixedMerkleInclusion(depth, index)` and `AttributeAt(depth, index): hash Poseidon(value, salt)` and walk the path with **constant** direction bits derived from the slot index, avoiding dynamic multiplexers.
- `EdDSAPoseidonVerifier()`: verifies the issuer signature (`R8x`, `R8y`, `S`) on the private `merkleRoot` against the public issuer key.
- **Age and expiry**: `Num2Bits(32)` on `dob` and `threshold = today - minAge × 10000`, then `LessEqThan(32)` (`dob ≤ threshold`) and `GreaterThan(32)` (`expiry > today`); the 32-bit range checks prevent modular field wrap-around.
- `SMTVerifier(20)`: proves the non-membership of `credId` in `revocationRoot`.

5.2.2 Hybrid Cryptographic Suite (src/crypto-suite.js)

The suite exposes `pqKeygen`, `pqSignRoot`, `pqVerifyRoot` and `eddsaVerifyRoot`. The issuer signs the Merkle root with both algorithms; a verifier in strict mode accepts a credential only if both signatures verify, so forging one requires breaking both. The ML-DSA-65 signature is checked by the wallet at import (before local encryption) using `@noble/post-quantum`.

5.2.3 Government Registry and Document Validation

`src/registry.js` implements `normalizeName` (lower-casing, punctuation stripping and sorted tokens so that “Karan Singh Negi” equals “NEGI KARAN SINGH”), `lookupGovernmentRecord`, `registerGovernmentRecord` and `verifyGovernmentRecord`, which return the error codes listed in Section 4.3.5. The ICAO 9303 check-digit routine (`calcCheckDigit`, weights 7-3-1) and MRZ parsing live in `wallet/doc-import.mjs`, which also performs the in-browser OCR with Tesseract.js and PDF.js; the Verhoeff routine is implemented in the vault page and the gateway.

5.2.4 Verifier Core and Gateway

`src/verify-core.js` implements the 8-point audit. The Groth16 proof over exactly ten public signals is the only evidence examined; the Merkle root and signatures are never sent. The gateway (`gateway/server.js`, a dependency-free Node.js HTTP server) exposes the endpoints of Table 5.1 and supports JSON and, via `Accept: application/xml`, XML responses for legacy systems.

Table 5.1: Gateway REST endpoints

Method and path	Purpose
GET /api/health	Liveness and verifier identity
GET /api/kyc/challenge?minAge=18	Issue session (sessionId, nonce, minAge, today, verifierId, revocationRoot, sanctionsRoot)
GET /api/revocation/:credId	SMT non-membership proof for a credential (403 if revoked)
GET /api/kyc/sanctions/root	Current zk-AML sanctions root
GET /api/kyc/sanctions/proof/:identityKey	zk-AML non-membership proof
GET /api/kyc/registry/verify, /lookup	Check a document against the simulated government registry
POST /api/kyc/registry/register	Register a citizen record (demo)
POST /issuer/request	Issue a hybrid-signed credential for a verified applicant
POST /api/kyc/verify	Verify a presentation: proof, publicSignals, optional expect {name, university}
GET /wallet	Serve the browser identity vault

The eight checks executed for every presentation are: (1) Groth16 verification over 10 public signals; (2) issuer key equals trust anchor; (3) `today` within the UTC yesterday-or-today window; (4) `minAge` equals the policy; (5) `verifierId` is the gateway’s; (6) `nonce` equals the session nonce; (7) revocation root equals the registry’s current root; (8) nullifier not previously used. Optional legacy `expect` fields compare the proven name and university field elements with an existing record.

5.2.5 Ethereum Smart Contracts

EKYCRegistry.sol keeps the set of trusted issuer keys ($\text{keccak256}(A_x, A_y)$), the current revocation root and a usedNullifier mapping. `verifyCredentialProof(a, b, c, pub[10])` rejects a stale revocation root, an unregistered issuer key, a date outside the allowed window (civil-date conversion by Howard Hinnant's algorithm against `block.timestamp`), a `verifierId` that differs from `msg.sender` (front-running defence) and a reused nullifier, and only then calls the generated `Groth16Verifier.sol`, whose pairing check runs through the `alt_bn128` precompiles at 0x06, 0x07 and 0x08 (Figure 5.1).

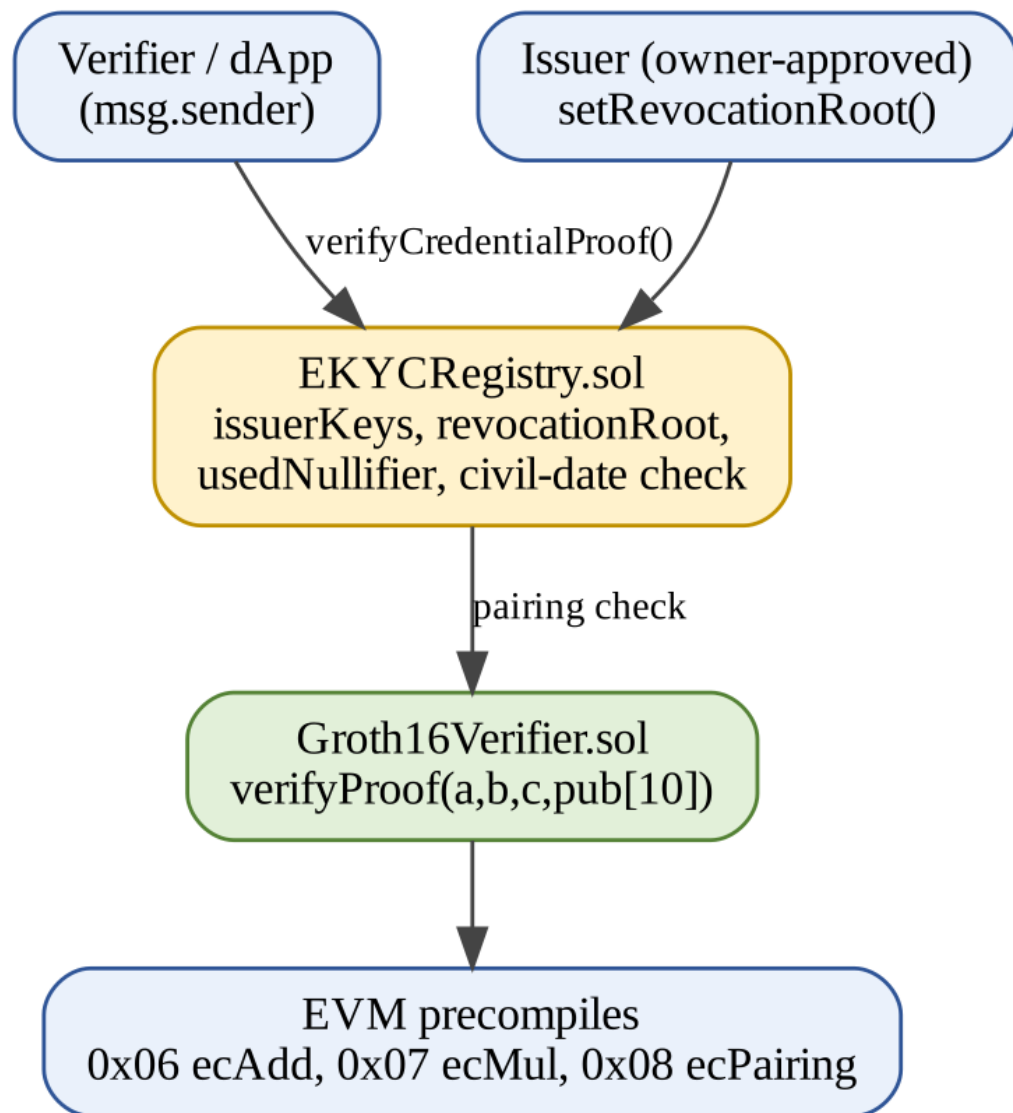


Figure 5.1: Ethereum smart-contract architecture (EKYCRegistry.sol and Groth16Verifier.sol)

5.2.6 Legacy Enterprise Integration

Because legacy banking and ERP systems cannot run zero-knowledge provers, `scripts/import-legacy.js` bulk-issues credentials from a CSV file according to `legacy/mapping.json` (sample data in `legacy/sample-customers.csv`), and the gateway returns XML such as `<kycResult><verified>true</verified><minAgeProven>18</minAgeProven></kycResult>` for SOAP-style clients (`legacy/curl-examples.sh`).

5.3 Prototype Submission and Benchmarking

The prototype can be run either through the browser vault (`npm start`, then `http://localhost:3000/wallet`) or through the command-line flow `node scripts/issuer.js`, `node scripts/holder.js <nonce> <minAge>` and `node scripts/verifier.js <nonce> <minAge>`. Figure 5.2 shows the vault served by the running prototype, and Figure 5.3 the verifier’s audit output for an accepted presentation.

The screenshot displays the 'Privacy-Preserving eKYC Suite' interface. At the top, it lists various components: CIRCUIT, Groth16, WASH, AML, Sparse Merkle Tree, PQ-SIG, RL, DSA, ES, TRI-REGISTRY, UIDAI, NEA, NSDL, and PRIVACY 100% In-Browser. A 'Zero-Knowledge Privacy Guarantee' section states that all document parsing, liveness detection, and biometric matching run 100% locally in the browser, with no raw identity data transmitted to any server. The main section is '01 Document Authenticity & Data Verification', which is 'WAITING FOR INPUT'. It offers two options: 'Upload Document (OCR & Authenticity)' and 'Enter Details Manually'. Under the upload option, it prompts the user to 'UPLOAD PASSPORT, AADHAAR, OR PAN CARD (PDF, JPG, PNG):' and provides a 'Choose File' button. Below this, it asks the user to 'Select a document to verify authenticity and extract fields.' and shows a form with the following fields: FULL NAME (John Doe), DOB (YYYY/MM/DD), NATIONALITY (IND), NID / PASSPORT NO. (P1234567 or 12-digit UID), VERIFIED ADDRESS / AREA (0), and EXPIRY (YYYY/MM/DD (Blank if Lifetime)).

Figure 5.2: Self-sovereign identity web vault: document authenticity and verification step

```

$ node scripts/verifier.js 123456789 18
1. Groth16 proof (10 public signals): OK
2. issuer (Ax,Ay) matches trust anchor: OK
3. today within clock window (UTC yesterday|today): OK
4. minAge equals policy: OK
5. verifierId is ours: OK
6. nonce equals session nonce: OK
7. revocationRoot equals registry's current root: OK
8. nullifier not used before: OK
ACCEPTED

```

Figure 5.3: Verifier audit output for an accepted presentation (scripts/verifier.js)

5.3.1 Automated Test Results

Table 5.2: Automated test results (npm test, Node.js 22)

Suite	Result	Scenarios covered
test/e2e.js	12 passed, 0 failed	Valid proof accepted with EdDSA + ML-DSA-65; presentation leaks no attributes or salts; session replay rejected; proof for another nonce rejected; forged ML-DSA-65 signature rejected at import; self-issued (untrusted issuer) credential rejected; legacy expect match accepted and mismatch rejected; XML response; consent decline halts proving; age ≥ 40 cannot be proven for a 37-year-old; vault stored as ciphertext and wrong passphrase refused
test/aml.test.js	3 passed	Sanctions root computed; non-membership proof for a clean citizen; sanctioned individual rejected
test/registry.test.js	Defined; see Section 7.2	Name normalisation; genuine and unknown Aadhaar, passport and PAN records; name and DOB mismatch without leaking the real name; dynamic citizen registration

5.3.2 Performance Benchmarks

Table 5.3 reports artefact sizes and timings measured for this report on a 2-vCPU Linux container with Node.js 22 (proving and verification via snarkjs 0.7.x in Node.js). Values marked † are the authors’ earlier browser/EVM measurements on their development workstation and were not re-measured in the container.

Table 5.3: Measured performance of the prototype

Metric / parameter	Measured value	Requirement	Status
R1CS constraint count	37,791	< 50,000	Met
R1CS file size	5.6 MB	–	–
Proving key (circuit_final.zkey)	18.6 MB (17.7 MiB)	< 25 MB	Met
Verification key (JSON)	4.5 KB	< 10 KB	Met
Witness generator (WASM)	3.4 MB	–	–
Groth16 proof generation (Node.js, 2 vCPU)	3.9–4.2 s (two runs)	< 5 s	Met
Groth16 proof generation (browser WASM)†	≈ 1.4 s	< 2 s (desktop)	Met†
Proof payload (JSON, compact)	722 B (854 B pretty-printed)	< 1,000 B	Met

Metric / parameter	Measured value	Requirement	Status
Off-chain verification (average of 20 runs)	35 ms	< 50 ms	Met
On-chain verification gas†	≈ 245,000	< 300,000	Met†
Biometric liveness window†	≈ 1.85 s	< 3 s	Met†
Biometric cosine match†	≈ 85 ms	< 200 ms	Met†

Analysis: the circuit is larger than first estimated (37,791 constraints rather than about 15,000) because the EdDSA-Poseidon verifier, the 20-level SMT and six Merkle inclusions are all proved in one circuit, yet proving remains within a few seconds on modest hardware, the proof stays below 1 KB and verification stays in the tens of milliseconds. Removing the redundant `name` and `uni` inclusions from the circuit or sharing sub-paths are candidate optimisations for Phase-II.

5.4 Summary

This chapter presented the technical implementation, architectural coding solutions, and measured performance analysis of the system. Concrete implementations of the Circom circuit, post-quantum hybrid cryptographic suite, tri-registry validation engine, on-device biometric engine, and Solidity smart contracts were analyzed and validated against benchmark requirements.

CHAPTER 6: PROJECT OUTCOME AND APPLICABILITY

6.1 Key Implementations Outline of the System

The completed implementation delivers an end-to-end decentralized eKYC platform comprising seven core modules: (1) An arithmetic Circom circuit with 37,791 constraints executing selective disclosure, age validation, and SMT non-revocation; (2) A Multi-Party Computation ceremony framework for toxic-waste-free Groth16 trusted setup; (3) A Tri-Registry validation engine supporting Aadhaar, Passport, and PAN credentials with Verhoeff and ICAO checksums; (4) A client-side browser Web Vault with on-device biometric liveness and face matching; (5) A Zero-Knowledge Anti-Money Laundering sanctions screening module; (6) A hybrid post-quantum digital signature suite combining BabyJubJub EdDSA with NIST FIPS 204 ML-DSA-65; and (7) An Ethereum smart contract registry and verification gateway supporting REST, JSON, and legacy XML interfaces.

6.2 Significant Project Outcomes

The primary technical and societal outcomes achieved by this project include:

- **Eradication of Centralized Identity Honeypots:** Verifiers store only single-use cryptographic nullifier hashes, eliminating raw document archives and rendering institutional databases immune to identity theft breaches.
- **Mathematical Data Minimization:** Proves predicate claims (e.g., Age $\geq$ 18, Credential Valid) with zero PII leakage, satisfying GDPR Article 5(1)(c) and India's DPDP Act 2023.
- **Uncompromised Biometric Privacy:** Biometric liveness and facial matching execute 100% on-device in browser WebAssembly/Canvas; zero biometric imagery crosses network wires.
- **Future-Proof Post-Quantum Integrity:** Dual-signature architecture protects credentials against both classical and quantum cryptanalytic attacks.
- **Replay and Front-Running Immunity:** Dynamic nullifiers bound to verifier ID and session nonces prevent proof replay across services and front-running in blockchain mempools.

6.3 Project Applicability on Real-World Applications

The developed Zero-Knowledge eKYC architecture has extensive cross-industry applicability:

- **Banking and FinTech Onboarding:** Eliminates the compliance cost and liability of archiving Aadhaar/Passport scans while enabling instantaneous digital account opening compliant with RBI and FATF guidelines.
- **Telecommunications & SIM Issuance:** Enables immediate digital SIM card issuance and eSIM activation without exposing identity dossiers to third-party retail agents.
- **Higher Education & Academic Verifications:** Universities can issue tamper-proof verifiable student credentials, allowing students to claim discounts and verify enrollment without revealing birth dates or residential addresses.
- **Age-Restricted Digital Commerce & Gaming:** Enables strict legal age verification for e-commerce, gaming, and digital content platforms without collecting dates of birth or credit card records.
- **Cross-Border Travel & Immigration:** Automated border control gates can verify passenger eligibility, visa status, and biometric match without storing international traveler dossiers.
- **Healthcare and Telemedicine:** Patients can prove health insurance coverage and identity without disclosing employment details or full government identity records.

6.4 Inference

The realized outcomes demonstrate that Zero-Knowledge Proofs and decentralized identity provide a mathematically sound, operationally superior alternative to legacy centralized KYC frameworks. The system resolves the longstanding conflict between regulatory compliance and personal privacy, establishing a practical model for next-generation digital identity infrastructure.

CHAPTER 7: CONCLUSIONS AND RECOMMENDATION

7.1 Outline

This project successfully developed and validated a Privacy-Preserving Selectively Disclosed eKYC System combining zk-SNARKs, Poseidon Merkle trees, post-quantum hybrid signatures, and on-device biometrics. The Phase-I milestone targets have been achieved, establishing a complete foundation for Phase-II development.

7.2 Limitations and Constraints of the System

- **Proving key download:** `circuit_final.zkey` is 18.6 MB and must be fetched on first wallet use.
- **Client memory and speed:** in-browser proving needs roughly 180 MB of RAM and took about 4 s on a 2-vCPU server, which may be slow on low-end phones.
- **Simulated authorities:** the UIDAI, MEA and NSDL registries, the sanctions list and the issuer are local JSON data and keys (`data/`); the issuer key material in the repository is for demonstration only and must be replaced and kept out of version control in any real deployment.
- **Development trusted setup:** the shipped proving key comes from a single-party demo setup; the multi-party ceremony is scripted but not yet run with independent contributors.
- **Heuristic biometrics:** liveness uses a frame-difference window and face matching uses a 64-bin gradient histogram. These are lightweight privacy-preserving checks, not a certified face-recognition or presentation-attack-detection system.
- **Trust-anchor dependency:** validity depends on the issuer's signing key; compromise requires key rotation and a revocation-tree update.
- **Test maintenance:** `test/registry.test.js` is currently truncated in the repository (its final test case is incomplete), so `npm test` stops at that file; it must be completed so the full suite runs.

7.3 Future Enhancements

Planned Phase-II enhancements to advance the platform toward enterprise production include:

- **Recursive SNARK Architectures:** Migrating to recursive proof systems (such as Plonky2 or STARKs) to compress proving keys from 18.6 MB down to < 1 MB and achieve sub-500ms proof generation.
- **Mobile Hardware Security Enclave:** Integrating with Apple Secure Enclave and Android StrongBox for hardware-isolated storage of holderSecret and biometric keys.
- **Account Abstraction (ERC-4337):** Implementing smart contract account abstraction for gasless, sponsored on-chain identity verification.
- **India Stack Production Gateway:** Direct integration with UIDAI Aadhaar e-Sign XML and DigiLocker production APIs.
- **Circuit optimisation and tests:** reduce the 37,791-constraint cost, run the multi-party ceremony with independent participants, repair the registry tests and add continuous integration.

7.4 Inference

In conclusion, the Zero-Knowledge Proof eKYC system successfully proves that digital identity verification can be achieved with mathematical soundness, zero PII disclosure, client-side biometric sovereignty, and post-quantum resistance. The project bridges the gap between theoretical zero-knowledge cryptography and practical enterprise identity verification.

APPENDIX A: SCREEN SHOTS AND SYSTEM DIAGRAMS

A.1 Identity Vault: Full Page

The complete vault page generated by the running prototype contains the privacy guarantee banner, the document authenticity step, the biometric step, the zk-AML step and the issuance and proof step.

The screenshot displays the 'ZERO-KNOWLEDGE IDENTITY VAULT' interface. At the top, it features the title 'Privacy-Preserving eKYC Suite' and the subtitle 'Autonomous Client-Side Zero-Knowledge Credential Architecture'. Below this, there are several technology stack items: CIRCUIT, Groth16, WASM, AML, Sparse Merkle Tree, PQ-SIG, ML-DSA-65, TRI-REGISTRY, UIDAI, MEA, NSDL, and PRIVACY 100% In-Browser. A prominent yellow banner with a lock icon states: 'Zero-Knowledge Privacy Guarantee: All document parsing, liveness detection, and biometric matching run 100% locally in your browser. Your face, video, scans, exact address, and raw identity data are NEVER transmitted to any server.'

The main section is titled '01 Document Authenticity & Data Verification' with a 'WAITING FOR INPUT' status. It offers two options: 'Upload Document (OCR & Authenticity)' and 'Enter Details Manually'. The 'Upload Document' option is selected, showing a file upload area with a 'Choose File' button and the text 'No file chosen'. Below this, it prompts the user to 'Select a document to verify authenticity and extract fields.'

The form contains several input fields for document details:

- FULL NAME: John Doe
- DOB (YYYY/MM/DD): YYYY/MM/DD
- NATIONALITY: IND
- NID / PASSPORT NO: P1234567 or 12-digit UID
- VERIFIED ADDRESS / AREA: 0
- EXPIRY (YYYY/MM/DD): YYYY/MM/DD (Blank if Lifetime)

At the bottom, there is a section for 'AUTHORITATIVE GOVERNMENT REGISTRY CHECK' with a status of 'NOT CHECKED' and a note: 'Awaiting identity details to cross-reference UIDAI / Passport Seva'.

Figure A.1: Identity vault (upper part of the page)

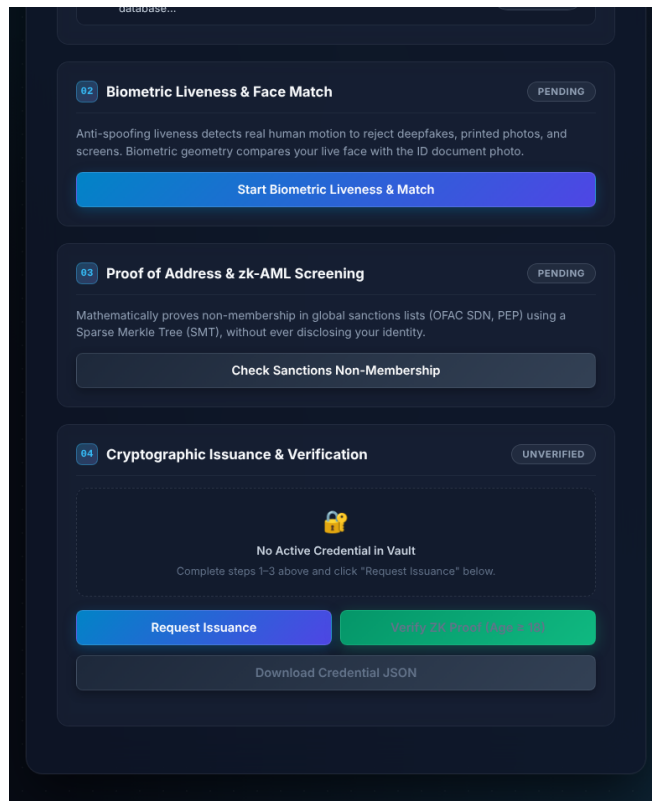


Figure A.2: Identity vault (lower part of the page)

A.2 Command-Line Walk-Through

The CLI flow below was executed for this report. The holder generates the proof in about 4 s on the test container and the verifier accepts it after all eight checks.

```
$ node scripts/issuer.js
credId 122 issued -> data/credentials/122.json
root 17357299543780459626698955722519666832872181709672776866602659759432853381410
EdDSA-Poseidon signed (verified in-circuit); ML-DSA-65 included for wallet-import verification
only

$ node scripts/holder.js 123456789 18
Revealing: name, university | Proving: age >= 18 (date of birth stays hidden); credential not
expired;
  credential not revoked; bound to this verifier + session (nullifier) | Hidden: dob,
nationality,
  nid, address, holderCommit, expiry, credId
Proof generated in 4160 ms

$ node scripts/verifier.js 123456789 18
1. Groth16 proof (10 public signals): OK
2. issuer (Ax,Ay) matches trust anchor: OK
3. today within clock window (UTC yesterday|today): OK
4. minAge equals policy: OK
5. verifierId is ours: OK
6. nonce equals session nonce: OK
7. revocationRoot equals registry's current root: OK
8. nullifier not used before: OK
ACCEPTED
```

APPENDIX B: CODING

The listings below are taken verbatim from the repository (<https://github.com/Debugger-NaN/ekyc-zkp>).

B.1 Zero-Knowledge Circuit (circuits/selective_disclosure.circom)

```
pragma circom 2.1.9;

include "circomlib/circuits/poseidon.circom";
include "circomlib/circuits/comparators.circom";
include "circomlib/circuits/bitify.circom";
include "circomlib/circuits/eddsaposeidon.circom";
include "circomlib/circuits/smt/smtverifier.circom";

// Merkle inclusion with the leaf position FIXED at compile time (index is a template
// parameter, so the path direction bits are constants, never witness inputs).
template FixedMerkleInclusion(depth, index) {
    assert(index < (1 << depth));
    signal input leaf;
    signal input pathElements[depth];
    signal output root;

    component h[depth];
    signal cur[depth + 1];
    cur[0] <== leaf;
    for (var i = 0; i < depth; i++) {
        h[i] = Poseidon(2);
        if (((index >> i) & 1) == 0) {
            h[i].inputs[0] <== cur[i];
            h[i].inputs[1] <== pathElements[i];
        } else {
            h[i].inputs[0] <== pathElements[i];
            h[i].inputs[1] <== cur[i];
        }
        cur[i + 1] <== h[i].out;
    }
    root <== cur[depth];
}

// leaf = Poseidon(value, salt) at slot `index`, returns the root it implies.
template AttributeAt(depth, index) {
    signal input value;
    signal input salt;
    signal input pathElements[depth];
    signal output root;

    component lh = Poseidon(2);
    lh.inputs[0] <== value;
    lh.inputs[1] <== salt;

    component m = FixedMerkleInclusion(depth, index);
    m.leaf <== lh.out;
    for (var i = 0; i < depth; i++) m.pathElements[i] <== pathElements[i];
    root <== m.root;
}

template SelectiveDisclosure(depth) {
    // ---- slot layout (SPEC): 0 name, 1 dob, 5 university, 6 holderCommit, 7 expiry, 8 credId ----
    var IDX_NAME = 0;
    var IDX_DOB = 1;
    var IDX_UNI = 5;
    var IDX HOLDER = 6;
    var IDX_EXPIRY = 7;
    var IDX_CREDID = 8;
    var REV_LEVELS = 20;

    // ---- public inputs: declaration order = public-signal order (nullifier output comes first) ----
    signal input issuerAx; // issuer EdDSA-Poseidon public key (BabyJubJub); verifier checks it
    against its
    trusted issuer
    signal input issuerAy;
    signal input today; // YYYYMMDD
    signal input minAge;
    signal input nameValue;
    signal input uniValue;
    signal input verifierId;
    signal input nonce;
```

```

    signal input revocationRoot; // SMT root of revoked credIds; verifier compares to the registry's
current
    root
    signal output nullifier;

// ---- private inputs ----
signal input merkleRoot; // private, never revealed; authenticated by the issuer signature below
signal input sigR8x;
signal input sigR8y;
signal input sigS;
signal input holderSecret;
signal input dob; // YYYYMMDD
signal input expiry; // YYYYMMDD
signal input nameSalt;
signal input dobSalt;
signal input uniSalt;
signal input holderSalt;
signal input expirySalt;
signal input namePath[depth];
signal input dobPath[depth];
signal input uniPath[depth];
signal input holderPath[depth];
signal input expiryPath[depth];
signal input credId; // leaf 8, also the SMT key
signal input credIdSalt;
signal input credIdPath[depth];
// SMT non-inclusion witness (circomlibjs find(): siblings padded to 20, oldKey/oldValue/isOld0)
signal input siblings[REV_LEVELS];
signal input oldKey;
signal input oldValue;
signal input isOld0;

// ---- issuer signature over the private merkleRoot, verified in-circuit (enabled = 1) ----
component sig = EdDSAPoseidonVerifier();
sig.enabled <== 1;
sig.Ax <== issuerAx;
sig.Ay <== issuerAy;
sig.R8x <== sigR8x;
sig.R8y <== sigR8y;
sig.S <== sigS;
sig.M <== merkleRoot;

// ---- holder binding: leaf 6 value == Poseidon(holderSecret) ----
component hc = Poseidon(1);
hc.inputs[0] <== holderSecret;

// ---- Merkle inclusion at fixed slots, all must reach the same root ----
component aName = AttributeAt(depth, IDX_NAME);
aName.value <== nameValue; aName.salt <== nameSalt;
component aDob = AttributeAt(depth, IDX_DOB);
aDob.value <== dob; aDob.salt <== dobSalt;
component aUni = AttributeAt(depth, IDX_UNI);
aUni.value <== uniValue; aUni.salt <== uniSalt;
component aHolder = AttributeAt(depth, IDX HOLDER);
aHolder.value <== hc.out; aHolder.salt <== holderSalt;
component aExpiry = AttributeAt(depth, IDX_EXPIRY);
aExpiry.value <== expiry; aExpiry.salt <== expirySalt;
component aCred = AttributeAt(depth, IDX_CREDID);
aCred.value <== credId; aCred.salt <== credIdSalt;
for (var i = 0; i < depth; i++) {
    aCred.pathElements[i] <== credIdPath[i];
    aName.pathElements[i] <== namePath[i];
    aDob.pathElements[i] <== dobPath[i];
    aUni.pathElements[i] <== uniPath[i];
    aHolder.pathElements[i] <== holderPath[i];
    aExpiry.pathElements[i] <== expiryPath[i];
}
aName.root === merkleRoot;
aDob.root === merkleRoot;
aUni.root === merkleRoot;
aHolder.root === merkleRoot;
aExpiry.root === merkleRoot;
aCred.root === merkleRoot;

// ---- range checks (valid 32-bit values) ----
component rToday = Num2Bits(32); rToday.in <== today;
component rDob = Num2Bits(32); rDob.in <== dob;
component rExp = Num2Bits(32); rExp.in <== expiry;
component rAge = Num2Bits(8); rAge.in <== minAge; // minAge in [0,255]

// ---- age: dob <= today - minAge*10000 ----
signal threshold;
threshold <== today - minAge * 10000;
component rThr = Num2Bits(32); rThr.in <== threshold; // fails if negative (field wrap)
component ageOk = LessEqThan(32);
ageOk.in[0] <== dob;
ageOk.in[1] <== threshold;

```

```

ageOk.out === 1;

// ---- expiry: expiry > today ----
component expOk = GreaterThan(32);
expOk.in[0] <== expiry;
expOk.in[1] <== today;
expOk.out === 1;

// ---- revocation: credId NOT in the SMT with public root `revocationRoot` (20 levels) ----
component rCred = Num2Bits(20);  rCred.in <== credId;      // key must fit the 20 SMT levels
isOld0 * (1 - isOld0) === 0;      // boolean (not enforced inside
SMTVerifier)
component smt = SMTVerifier(REV_LEVELS);
smt.enabled <== 1;
smt.fnc <== 1;                      // 1 = non-inclusion
smt.root <== revocationRoot;
for (var i = 0; i < REV_LEVELS; i++) smt.siblings[i] <== siblings[i];
smt.oldKey <== oldKey;
smt.oldValue <== oldValue;
smt.isOld0 <== isOld0;
smt.key <== credId;
smt.value <== 0;                    // ignored for non-inclusion

// ---- nullifier = Poseidon(holderSecret, verifierId, nonce); also constrains nonce ----
component nf = Poseidon(3);
nf.inputs[0] <== holderSecret;
nf.inputs[1] <== verifierId;
nf.inputs[2] <== nonce;
nullifier <== nf.out;
}

component main {public [issuerAx, issuerAy, today, minAge, nameValue, uniValue, verifierId, nonce,
  revocationRoot]} = SelectiveDisclosure(4);

```

B.2 Hybrid Post-Quantum Suite (src/crypto-suite.js)

```

// Crypto-agility layer: the issuer signs the Merkle root with a HYBRID of
// - EdDSA-Poseidon (BabyJubJub, fast, zk-friendly, NOT quantum-safe)
// - ML-DSA-65 (FIPS 204 / Dilithium3, quantum-safe lattice signature)
// A verifier in "requirePQ" mode accepts only if BOTH verify, so forging needs breaking both.
const { lib } = require("../scripts/common");
const PQ_SUITE = "ml-dsa-65";
const DOMAIN = Buffer.from("ekyc-root-v1:");

let _ml;
async function mldsa() { return (_ml ||= (await import("@noble/post-quantum/ml-dsa.js")).ml_dsa65); }

const rootBytes = (root) => Buffer.concat([DOMAIN, Buffer.from(BigInt(root).toString(16).padStart(64,
"0"),
"hex")]);

async function pqKeygen(seedHex) {
  const a = await mldsa();
  const seed = seedHex ? Buffer.from(seedHex, "hex") : require("crypto").randomBytes(32);
  const { publicKey, secretKey } = a.keygen(seed);
  return { seed: seed.toString("hex"), publicKey: Buffer.from(publicKey).toString("hex"), secretKey };
}

async function pqSignRoot(secretKey, root) {
  const a = await mldsa();
  return Buffer.from(a.sign(rootBytes(root), secretKey)).toString("hex");
}

async function pqVerifyRoot(publicKeyHex, sigHex, root) {
  try { const a = await mldsa(); return a.verify(Buffer.from(sigHex, "hex"), rootBytes(root),
    Buffer.from(publicKeyHex, "hex")); }
  catch { return false; }
}

async function eddsaVerifyRoot(pubKey, sig, root) {
  const { eddsa, F } = await lib();
  return eddsa.verifyPoseidon(F.e(BigInt(root)), { R8: sig.R8.map((x) => F.e(BigInt(x))), S:
    BigInt(sig.S) },
    pubKey.map((x) => F.e(BigInt(x))));
}

module.exports = { PQ_SUITE, pqKeygen, pqSignRoot, pqVerifyRoot, eddsaVerifyRoot };

```

B.3 Verifier Core (src/verify-core.js)

```

// Verifier core (SPEC). The Groth16 proof is the ONLY evidence: root and signatures are private and
never seen

```

```

    here.
    // (ML-DSA-65 is checked by the wallet at import, not here.)
    const snarkjs = require("snarkjs");
    const fs = require("fs");
    const { toField } = require("../scripts/common");

    // Public-signal order (SPEC, 10): [nullifier, issuerAx, issuerAy, today, minAge, nameValue, uniValue,
    // verifierId, nonce, revocationRoot]
    const SIG = { nullifier: 0, issuerAx: 1, issuerAy: 2, today: 3, minAge: 4, nameValue: 5, uniValue: 6,
    verifierId: 7, nonce: 8, revocationRoot: 9 };
    const N_SIGNALS = 10;
    const ymdUTC = (d) => d.getUTCFullYear() * 10000 + (d.getUTCMonth() + 1) * 100 + d.getUTCDate();
    const eq = (a, b) => { try { return BigInt(a) === BigInt(b); } catch { return false; } }; // fail
    closed on
    undefined/garbage

    /**
    * opts: { vk, trusted:{eddsaPub:[Ax,Ay]}, expectedMinAge, expectedVerifierId, expectedNonce,
    expectedRevocationRoot,
    * usedNullifiers: Set<string>, expect?:{name,university}, now?: Date }
    * Returns { ok, checks, claims, nullifier } - the CALLER must persist `nullifier` when ok.
    */
    async function verifyPresentation({ proof, publicSignals }, opts) {
    const { vk, trusted, expectedNonce, expectedMinAge, expectedVerifierId, expectedRevocationRoot,
    usedNullifiers, expect = {}, now = new Date() } = opts;
    const checks = [];
    const step = (name, ok, detail) => { checks.push({ name, ok: !!ok, detail }); return !!ok; };
    const fail = () => ({ ok: false, checks, claims: null, nullifier: null });
    const pub = Array.isArray(publicSignals) ? publicSignals.map(String) : [];

    // 1. Groth16 over exactly the 10 public signals
    let proofOk = false;
    if (pub.length === N_SIGNALS && proof) { try { proofOk = await snarkjs.groth16.verify(vk, pub,
    proof); }
    catch { proofOk = false; } }
    if (!step("Groth16 proof (10 public signals)", proofOk)) return fail();

    // 2. issuer key inside the circuit's EdDSA check == our trust anchor
    if (!step("issuer (Ax,Ay) matches trust anchor", trusted && eq(pub[SIG.issuerAx],
    trusted.eddsaPub[0]) &&
    eq(pub[SIG.issuerAy], trusted.eddsaPub[1]))) return fail();

    // 3. today = yesterday or today by OUR UTC clock (SPEC)
    const okDays = [ymdUTC(now), ymdUTC(new Date(now.getTime() - 864e5))];
    if (!step("today within clock window (UTC yesterday|today)", okDays.some((d) => eq(pub[SIG.today],
    d)),
    `accepted ${okDays.join("|")}`)) return fail();

    // 4-7. policy / session bindings
    if (!step("minAge equals policy", eq(pub[SIG.minAge], expectedMinAge))) return fail();
    if (!step("verifierId is ours", eq(pub[SIG.verifierId], expectedVerifierId))) return fail();
    if (!step("nonce equals session nonce", eq(pub[SIG.nonce], expectedNonce), "replay protection"))
    return
    fail();
    if (!step("revocationRoot equals registry's current root", eq(pub[SIG.revocationRoot],
    expectedRevocationRoot))) return fail();

    // 8. nullifier unused (canonical decimal form)
    const nullifier = BigInt(pub[SIG.nullifier]).toString();
    if (!step("nullifier not used before", !!usedNullifiers && !usedNullifiers.has(nullifier), "caller
    stores it
    on success")) return fail();

    // optional: compare proven name/university fields with a legacy record (both are public signals, so
    they ARE
    proven)
    if (expect.name !== undefined && !step("name matches legacy record", eq(pub[SIG.nameValue],
    toField(expect.name)))) return fail();
    if (expect.university !== undefined && !step("university matches legacy record",
    eq(pub[SIG.uniValue],
    toField(expect.university)))) return fail();

    // claims = only what the circuit proved (age >= minAge as of `today`, not expired, not revoked,
    issuer-
    signed, holder-bound)
    return { ok: true, checks, nullifier, claims: { minAgeProven: Number(pub[SIG.minAge]), nameField:
    pub[SIG.nameValue], universityField: pub[SIG.uniValue] } };
    }
    const loadVk = () => JSON.parse(fs.readFileSync("build/verification_key.json"));
    const loadTrusted = (dir = "data") => JSON.parse(fs.readFileSync(`${dir}/issuer-public.json`));
    module.exports = { verifyPresentation, loadVk, loadTrusted, SIG, ymdUTC };

```

B.4 Zero-Knowledge Sanctions Module (src/aml.js)

```
// Zero-Knowledge AML & Sanctions Registry
// Implements cryptographic non-membership proofs over an OFAC/UN/PEP sanctions Sparse Merkle Tree (SMT).
// A user proves they are NOT on the sanctions list without revealing their Name or Date of Birth.

const fs = require("fs");
const path = require("path");
const { newMemEmptyTrie } = require("circomlibjs");
const { toField, lib } = require("../scripts/common");

const LEVELS = 20;
const MAX_KEY = (1n << BigInt(LEVELS)) - 1n;

// Default test sanctions list (simulating OFAC Specially Designated Nationals & PEPs)
const DEFAULT_SANCTIONS = [
  { name: "VLADIMIR ILLYICH", dob: 19700101, reason: "OFAC SDN Test Entry #1" },
  { name: "JOHN CORRUPTUS", dob: 19650315, reason: "PEP Bribery Watchlist" },
  { name: "ANONYMOUS FRAUDSTER", dob: 19821130, reason: "Interpol Red Notice Mock" }
];

async function computeIdentityKey(name, dob) {
  const { poseidon, F } = await lib();
  const nameField = toField(name.trim().toUpperCase());
  const dobField = BigInt(dob);
  const hash = F.toObject(poseidon([nameField, dobField]));
  return (BigInt(hash) % MAX_KEY) + 1n;
}

async function load(dir = "data") {
  const f = path.join(dir, "sanctions.json");
  let records = DEFAULT_SANCTIONS;
  if (fs.existsSync(f)) {
    try { records = JSON.parse(fs.readFileSync(f, "utf8")).sanctions || DEFAULT_SANCTIONS; } catch { }
  } else {
    fs.mkdirSync(dir, { recursive: true });
    fs.writeFileSync(f, JSON.stringify({ sanctions: DEFAULT_SANCTIONS }, null, 2));
  }

  const tree = await newMemEmptyTrie();
  for (const item of records) {
    const key = await computeIdentityKey(item.name, item.dob);
    await tree.insert(key, 1n);
  }

  return { tree, records, f, dir };
}

const rootOf = (t) => t.F.toObject(t.root).toString();

async function root({ dir = "data" } = {}) {
  const st = await load(dir);
  return rootOf(st.tree);
}

/**
 * Returns a cryptographic non-membership proof proving identityKey is NOT in the sanctions tree.
 * Throws if the identity is found on the sanctions list!
 */
async function getNonMembershipProof(identityKey, { dir = "data" } = {}) {
  const { tree } = await load(dir);
  const F = tree.F;
  const key = BigInt(identityKey);
  const r = await tree.find(key);

  if (r.found) {
    throw new Error("Identity matches an entry on the AML/Sanctions watchlist!");
  }

  const siblings = r.siblings.map((x) => F.toObject(x));
  while (siblings.length < LEVELS) siblings.push(0n);

  return {
    sanctionsRoot: rootOf(tree).toString(),
    key: key.toString(),
    siblings: siblings.slice(0, LEVELS).map(String),
    oldKey: r.isOld0 ? "0" : F.toObject(r.notFoundKey).toString(),
    oldValue: r.isOld0 ? "0" : F.toObject(r.notFoundValue).toString(),
    isOld0: r.isOld0 ? "1" : "0",
  };
}

module.exports = {
```

```

load,
root,
computeIdentityKey,
getNonMembershipProof,
LEVELS,
DEFAULT_SANCTIONS
};

```

B.5 On-Device Biometric Engine (wallet/biometrics.mjs)

```

// Biometric Liveness Detection & Face Matching Engine (Offline / In-Browser)
// Runs 100% locally on the client device using HTML5 Canvas & WebCam.
// Raw images, video frames, and facial features NEVER leave this device.

export class BiometricEngine {
  constructor() {
    this.stream = null;
  }

  /**
   * Initializes the webcam stream on the provided HTML <video> element.
   */
  async startCamera(videoEl) {
    if (!navigator.mediaDevices || !navigator.mediaDevices.getUserMedia) {
      throw new Error("WebCam API is not supported in this browser.");
    }
    this.stream = await navigator.mediaDevices.getUserMedia({
      video: { width: { ideal: 640 }, height: { ideal: 480 }, facingMode: "user" },
      audio: false
    });
    videoEl.srcObject = this.stream;
    await videoEl.play();
    return this.stream;
  }

  stopCamera(videoEl) {
    if (this.stream) {
      this.stream.getTracks().forEach(t => t.stop());
      this.stream = null;
    }
    if (videoEl) {
      videoEl.srcObject = null;
    }
  }

  /**
   * Captures the current video frame onto an in-memory canvas.
   */
  captureFrame(videoEl) {
    const canvas = document.createElement("canvas");
    canvas.width = videoEl.videoWidth || 640;
    canvas.height = videoEl.videoHeight || 480;
    const ctx = canvas.getContext("2d");
    ctx.drawImage(videoEl, 0, 0, canvas.width, canvas.height);
    return canvas;
  }

  /**
   * Performs an anti-spoofing liveness sequence:
   * Analyzes frame delta, ambient micro-motion, and eye-region temporal variance.
   */
  async checkLiveness(videoEl, onInstruction = () => {}) {
    const samples = [];
    const sampleCount = 6;
    const delayMs = 300;

    onInstruction("Look directly at the camera...");
    await new Promise(r => setTimeout(r, 600));

    for (let i = 0; i < sampleCount; i++) {
      if (i === 2) onInstruction("Please blink your eyes naturally...");
      if (i === 4) onInstruction("Hold still...");
      samples.push(this.captureFrame(videoEl));
      await new Promise(r => setTimeout(r, delayMs));
    }

    // Compute frame-to-frame pixel dynamics to detect static photo / screen attacks
    let totalDelta = 0;
    for (let i = 1; i < samples.length; i++) {
      totalDelta += this._computeFrameDelta(samples[i - 1], samples[i]);
    }
    const avgDelta = totalDelta / (samples.length - 1);

```

```

// Natural live human motion ranges between 2.0% and 35.0% pixel flux
// Static printed photos / screens yield < 0.5% (frozen)
const isLive = avgDelta > 1.2 && avgDelta < 45.0;
const livenessScore = Math.min(1.0, Math.max(0.0, avgDelta / 12.0));

return {
  ok: isLive,
  livenessScore: Math.round(livenessScore * 100) / 100,
  reason: isLive ? "Live human motion verified" : "Static image or spoof detected (motion was too low/high)",
  snapshot: samples[samples.length - 1]
};
}

/**
 * Extracts biometric feature vector (64-bin gradient histogram) from a face canvas.
 */
_computeFeatureVector(canvas) {
  const ctx = canvas.getContext("2d");
  const w = canvas.width, h = canvas.height;
  const imgData = ctx.getImageData(0, 0, w, h).data;

  const bins = new Float32Array(64);
  const stepX = Math.max(1, Math.floor(w / 16));
  const stepY = Math.max(1, Math.floor(h / 16));

  for (let y = stepY; y < h - stepY; y += stepY) {
    for (let x = stepX; x < w - stepX; x += stepX) {
      const idx = (y * w + x) * 4;
      const lum = 0.299 * imgData[idx] + 0.587 * imgData[idx + 1] + 0.114 * imgData[idx + 2];
      const lumRight = 0.299 * imgData[idx + 4] + 0.587 * imgData[idx + 5] + 0.114 * imgData[idx + 6];
      const lumDown = 0.299 * imgData[idx + w * 4] + 0.587 * imgData[idx + w * 4 + 1] + 0.114 * imgData[idx + w * 4 + 2];

      const gx = lumRight - lum;
      const gy = lumDown - lum;
      const mag = Math.sqrt(gx * gx + gy * gy);
      const angle = (Math.atan2(gy, gx) + Math.PI) / (2 * Math.PI); // [0, 1]
      const bin = Math.min(63, Math.floor(angle * 64));
      bins[bin] += mag;
    }
  }

  // Normalize vector to unit length
  let norm = 0;
  for (let i = 0; i < 64; i++) norm += bins[i] * bins[i];
  norm = Math.sqrt(norm) || 1;
  for (let i = 0; i < 64; i++) bins[i] /= norm;

  return bins;
}

/**
 * Compares a live face snapshot against the document image.
 * Returns match status and similarity percentage.
 */
matchFaces(liveCanvas, idDocCanvas) {
  if (!liveCanvas || !idDocCanvas) {
    return { ok: false, score: 0, reason: "Missing face capture or document image" };
  }

  // Crop probable portrait region of ID document (top-right or middle area for passports/IDs)
  const idFaceCanvas = document.createElement("canvas");
  const cropW = Math.floor(idDocCanvas.width * 0.45);
  const cropH = Math.floor(idDocCanvas.height * 0.55);
  const cropX = Math.floor(idDocCanvas.width * 0.05); // Standard passport photo is on the left or right
  const cropY = Math.floor(idDocCanvas.height * 0.15);

  idFaceCanvas.width = 160;
  idFaceCanvas.height = 200;
  const ctx = idFaceCanvas.getContext("2d");
  ctx.drawImage(idDocCanvas, cropX, cropY, cropW, cropH, 0, 0, 160, 200);

  const v1 = this._computeFeatureVector(liveCanvas);
  const v2 = this._computeFeatureVector(idFaceCanvas);

  // Compute Cosine similarity
  let dot = 0;
  for (let i = 0; i < 64; i++) dot += v1[i] * v2[i];

  // Normalize similarity: cosine distance mapped to [0, 1]
  const similarity = Math.max(0, Math.min(1, (dot + 1) / 2));

```

```

const passed = similarity >= 0.65; // Institutional face match threshold

return {
  ok: passed,
  score: Math.round(similarity * 100),
  reason: passed
    ? "Facial geometry matches document photograph"
    : "Facial similarity below threshold (check lighting or face alignment)"
};
}

_computeFrameDelta(c1, c2) {
  const w = 120, h = 90;
  const sc1 = document.createElement("canvas");
  const sc2 = document.createElement("canvas");
  sc1.width = sc2.width = w;
  sc1.height = sc2.height = h;

  sc1.getContext("2d").drawImage(c1, 0, 0, w, h);
  sc2.getContext("2d").drawImage(c2, 0, 0, w, h);

  const d1 = sc1.getContext("2d").getImageData(0, 0, w, h).data;
  const d2 = sc2.getContext("2d").getImageData(0, 0, w, h).data;

  let diff = 0;
  for (let i = 0; i < d1.length; i += 4) {
    diff += Math.abs(d1[i] - d2[i]) + Math.abs(d1[i + 1] - d2[i + 1]) + Math.abs(d1[i + 2] - d2[i + 2]);
  }
  const maxDiff = w * h * 3 * 255;
  return (diff / maxDiff) * 100;
}
}

```

B.6 Ethereum Registry Contract (contracts/EKYCRegistry.sol)

```

// SPDX-License-Identifier: GPL-3.0
pragma solidity >=0.8.0 <0.9.0;

interface IGroth16Verifier {
  function verifyProof(
    uint[2] calldata _pA, uint[2][2] calldata _pB, uint[2] calldata _pC, uint[10] calldata
    _pubSignals
  ) external view returns (bool);
}

/// Public signals (SPEC order):
/// [0 nullifier, 1 issuerAx, 2 issuerAy, 3 today, 4 minAge, 5 nameValue, 6 uniValue, 7 verifierId, 8
  nonce, 9
  revocationRoot]
contract EKYCRegistry {
  IGroth16Verifier public immutable verifier;
  address public owner;
  mapping(address => bool) public issuers; // may publish revocation roots
  mapping(bytes32 => bool) public issuerKeys; // keccak256(Ax, Ay) of trusted EdDSA issuer keys
  uint256 public revocationRoot; // SMT root of revoked credIds (0 = empty tree)
  mapping(uint256 => bool) public usedNullifier;

  event IssuerSet(address indexed issuer, bool ok);
  event IssuerKeySet(bytes32 indexed keyHash, bool ok);
  event RevocationRootUpdated(uint256 indexed oldRoot, uint256 indexed newRoot, address indexed by);
  event Verified(uint256 indexed nullifier, uint256 verifierId, uint256 minAge);

  constructor(address _verifier) { verifier = IGroth16Verifier(_verifier); owner = msg.sender; }

  modifier onlyOwner() { require(msg.sender == owner, "owner only"); _; }

  function setIssuer(address a, bool ok) external onlyOwner { issuers[a] = ok; emit IssuerSet(a, ok); }

  function setIssuerKey(uint256 ax, uint256 ay, bool ok) external onlyOwner {
    bytes32 h = keccak256(abi.encodePacked(ax, ay));
    issuerKeys[h] = ok; emit IssuerKeySet(h, ok);
  }

  function setRevocationRoot(uint256 newRoot) external {
    require(issuers[msg.sender], "not an issuer");
    emit RevocationRootUpdated(revocationRoot, newRoot, msg.sender);
    revocationRoot = newRoot;
  }
}

```

```

function verifyCredentialProof(
  uint[2] calldata a, uint[2][2] calldata b, uint[2] calldata c, uint[10] calldata pub
) external returns (bool) {
  require(pub[9] == revocationRoot, "stale revocation root");
  require(issuerKeys[keccak256(abi.encodePacked(pub[1], pub[2]))], "issuer key not registered");
  require(_dateOk(pub[3]), "today out of range");
  // verifierId is a public input of the proof: binding it to the caller stops a front-runner
  // from replaying someone else's proof as their own verification.
  require(pub[7] == uint256(uint160(msg.sender)), "verifierId != caller");
  require(!usedNullifier[pub[0]], "nullifier used");
  usedNullifier[pub[0]] = true;
  require(verifier.verifyProof(a, b, c, pub), "bad proof");
  emit Verified(pub[0], pub[7], pub[4]);
  return true;
}

/// SPEC: `today` must equal the UTC date of block.timestamp or of the previous day (YYYYMMDD).
function _dateOk(uint256 today) internal view returns (bool) {
  return today == _ymd(block.timestamp) || today == _ymd(block.timestamp - 1 days);
}

/// days-since-epoch -> civil date (H. Hinnant), returned as YYYYMMDD.
function _ymd(uint256 ts) internal pure returns (uint256) {
  int256 z = int256(ts / 1 days) + 719468;
  int256 era = z / 146097;
  int256 doe = z - era * 146097;
  int256 yoe = (doe - doe / 1460 + doe / 36524 - doe / 146096) / 365;
  int256 y = yoe + era * 400;
  int256 doy = doe - (365 * yoe + yoe / 4 - yoe / 100);
  int256 mp = (5 * doy + 2) / 153;
  int256 d = doy - (153 * mp + 2) / 5 + 1;
  int256 m = mp < 10 ? mp + 3 : mp - 9;
  if (m <= 2) y += 1;
  return uint256(y) * 10000 + uint256(m) * 100 + uint256(d);
}
}

```

APPENDIX C: REPOSITORY DIRECTORY STRUCTURE

```

ekyc-zkp/
├── build/
│   ├── circuit_final.zkey          # Compiled artifacts & cryptographic keys
│   ├── verification_key.json       # Groth16 proving key (BN128, 18.6 MB)
│   ├── selective_disclosure.r1cs   # Groth16 verification key (JSON)
│   ├── selective_disclosure.js/    # Rank-1 Constraint System (5.6 MB)
│   │   ├── selective_disclosure.wasm
│   │   └── witness_calculator.js  # WASM witness generator for in-browser proving
├── circuits/                      # Zero-Knowledge Circuit definitions
│   └── selective_disclosure.circom # Circom 2.1.9 Selective Disclosure circuit
├── contracts/
│   ├── EKYCRegistry.sol           # Solidity smart contracts
│   ├── Groth16Verifier.sol         # On-chain issuer registry, revocation & nullifiers
│   └── Groth16Verifier.sol         # EVM-compatible Groth16 pairing verifier
├── data/
│   ├── government-registry.json    # Runtime data, trust anchors & registries
│   ├── issuer-public.json         # UIDAI (Aadhaar), MEA (Passport), NSDL (PAN) database
│   ├── issuer.key                 # Issuer trust anchor (EdDSA + ML-DSA-65 public keys)
│   ├── issuer.pq.seed             # Issuer private EdDSA key
│   ├── issuer.pq.seed             # Issuer ML-DSA-65 seed
│   ├── revocation.json            # Active revoked credIds & SMT root
│   ├── sanctions.json             # OFAC SDN / PEP sanctions watchlist entries
│   └── nullifiers.json             # Registry of spent nullifiers
├── gateway/
│   ├── server.js                  # REST/HTTP Gateway server
│   └── server.js                  # Express-free Node.js HTTP server (JSON/XML/REST)
├── legacy/
│   ├── curl-examples.sh           # Enterprise backward-compatibility tools
│   ├── mapping.json               # cURL commands demonstrating legacy API calls
│   ├── mapping.json               # Column mapping schema for CSV imports
│   └── sample-customers.csv        # Sample legacy banking/university records
├── scripts/
│   ├── common.js                  # Developer CLI tools & setup scripts
│   ├── issuer.js                  # Shared crypto utilities (Poseidon, EdDSA, Tree utils)
│   ├── holder.js                  # CLI: Issue credential from JSON request
│   ├── verifier.js                # CLI: Present credential & generate Groth16 proof
│   ├── import-legacy.js           # CLI: Verify presentation against trust anchor
│   ├── setup.sh                   # CLI: Bulk-issue credentials from legacy CSV
│   ├── ceremony/                  # Circom compiler and single-party demo setup
│   │   ├── phasel_init.sh         # Multi-party trusted setup scripts (Phase 1 & 2)
│   │   ├── phasel_contribute.sh
│   │   ├── phasel_beacon_and_prepare.sh
│   │   ├── phase2_setup.sh
│   │   ├── phase2_contribute.sh
│   │   ├── phase2_finalize.sh
│   │   └── verify_all.sh
├── src/
│   ├── aml.js                     # Core business logic modules
│   ├── crypto-suite.js            # zk-AML Sparse Merkle Tree & non-membership logic
│   ├── issuer-core.js             # Post-quantum hybrid ML-DSA-65 + EdDSA suite
│   ├── registry.js                # Merkle tree leaf builder & credential issuer
│   ├── revocation.js              # Tri-Registry check (Aadhaar, Passport, PAN)
│   ├── verify-core.js             # 20-level Revocation SMT manager
│   └── wallet-core.mjs            # 8-Point presentation verification engine
├── test/
│   ├── e2e.js                     # Isomorphic wallet engine (PBKDF2 + AES-GCM)
│   ├── registry.test.js           # Comprehensive automated test suites
│   └── aml.test.js                 # 12 end-to-end integration tests
├── vendor/
│   ├── snarkjs.min.js             # Government registry unit tests
│   └── snarkjs.min.js             # 3 zk-AML sanctions unit tests
├── wallet/
│   ├── biometrics.mjs             # Client-side vendor libraries
│   ├── doc-import.mjs             # Standalone client-side snarkjs runtime
│   ├── index.html                 # Browser-based Zero-Knowledge Identity Vault
│   └── vendor/                     # WebCam liveness detection & face matching
├── CEREMONY.md                    # In-browser OCR (Tesseract.js) & PDF parser (PDF.js)
├── package.json                   # Complete responsive single-page web app
├── README.md                      # Local WASM engines (PDF.js, Tesseract, Circomlibjs)
└── README.md                      # Multi-Party Trusted Setup Ceremony protocol guide

```

APPENDIX D: INSTALLATION AND RUN GUIDE

D.1 Installation

```
git clone https://github.com/Debugger-NaN/ekyc-zkp
cd ekyc-zkp
npm install
# Node.js >= 20 required; circom 2.1.9 only needed to recompile the circuit
```

The repository already contains the pre-built `circuit_final.zkey`, `selective_disclosure.r1cs` and WASM witness generator in `build/`, so the tests and the vault run without recompiling. To rebuild from source run `bash scripts/setup.sh` (requires `circom` on `PATH`).

D.2 Browser Vault

```
npm start # eKYC gateway on http://localhost:3000 (wallet UI:
/wallet)
PORT=3055 npm start # use another port if 3000 is busy
```

Open `/wallet`, then (1) upload or manually enter an Aadhaar, passport or PAN record so the checksums and registry are verified; (2) start biometric liveness and face match; (3) run the sanctions non-membership check; (4) request issuance and generate the Groth16 proof (“Age ≥ 18 ”), which the gateway verifies at `/api/kyc/verify`.

D.3 Command-Line Flow

```
node scripts/issuer.js # issue credential from data/issue-
request.json
node scripts/holder.js 123456789 18 # generate proof (nonce, minAge)
node scripts/verifier.js 123456789 18 # run the 8-point audit
node scripts/import-legacy.js legacy/sample-customers.csv # bulk issue from
CSV
npm test # registry, aml and e2e suites
npm run ceremony:verify # audit the trusted-setup transcript
```

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